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Mid term file Zoo 506

Subjective/highlighted lectures

By M@ry@m S@leem kh@w@j@

"The Expert Zoologist"

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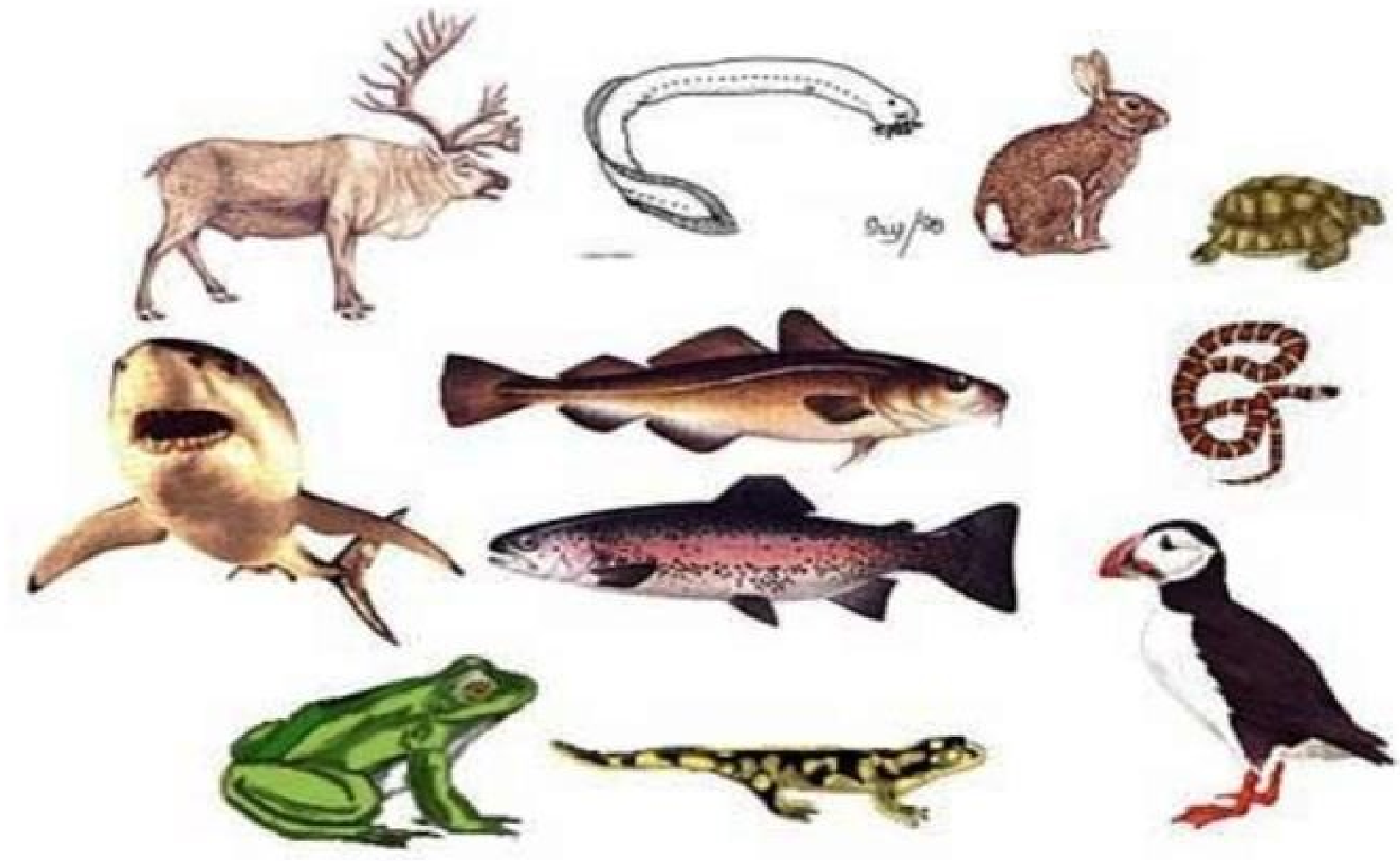
1-Relationship of insects with other animals Groups of animals

Q:name The groups of animals

- First group:The larger groups of animals known by science known as the Chordata e.g. mammals, birds, reptiles and fishes,



- ***Second group*** *less well known Another group, called the* ***Mollusca****, e.g. the snails, clams, etc*

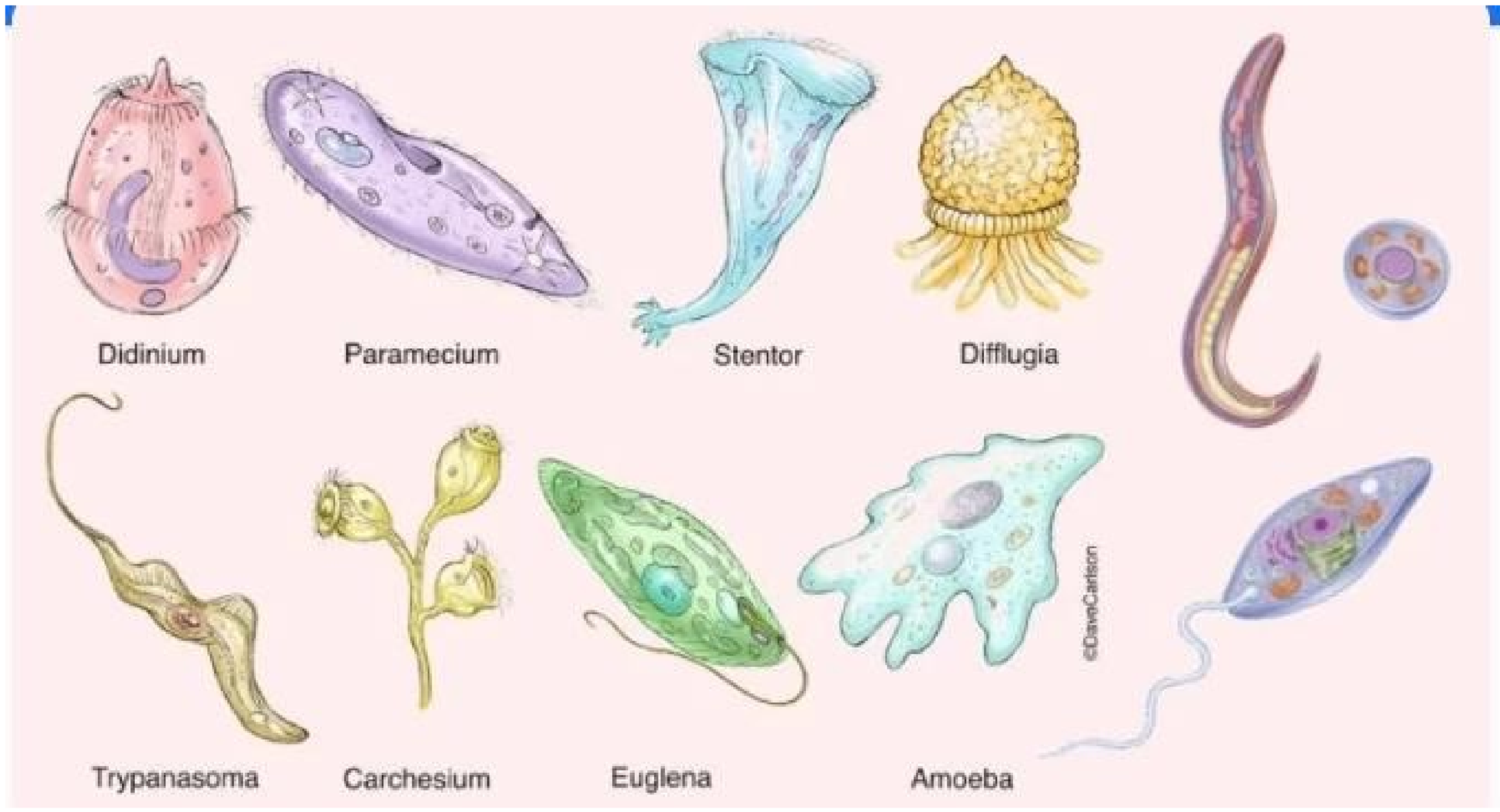
ELK
SLUG
OYSTER
SQUID
OCTOPUS
AMMONITES
CUTTLEFISH
MUSSELS
NAUTILIDAE
SCALLOPS
SEA SLUGS
SEA SNAILS
NUDIBRANCH
BUCCINIDAE
CLAMS

. – ***Third group***

The starfish and sea urchin, often seen at the seashore belongs with ***Echinodermata****.*

– ***Fourth group***

Tiny beings too small to be seen without the help of a microscope, are included 'in the group Protozoa



- ***Fifth group:***

Mainly of soft, jellylike animals, the more common larger members being called jellyfish,
and to this group the name
Coelenterata/cnidarians

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Q: describe arthropods/phylum arthropoda?

Arthropoda o *Several other groups including various kinds of worms and less familiar forms are also known Arthropoda*

o *The largest group of all, however, is the*

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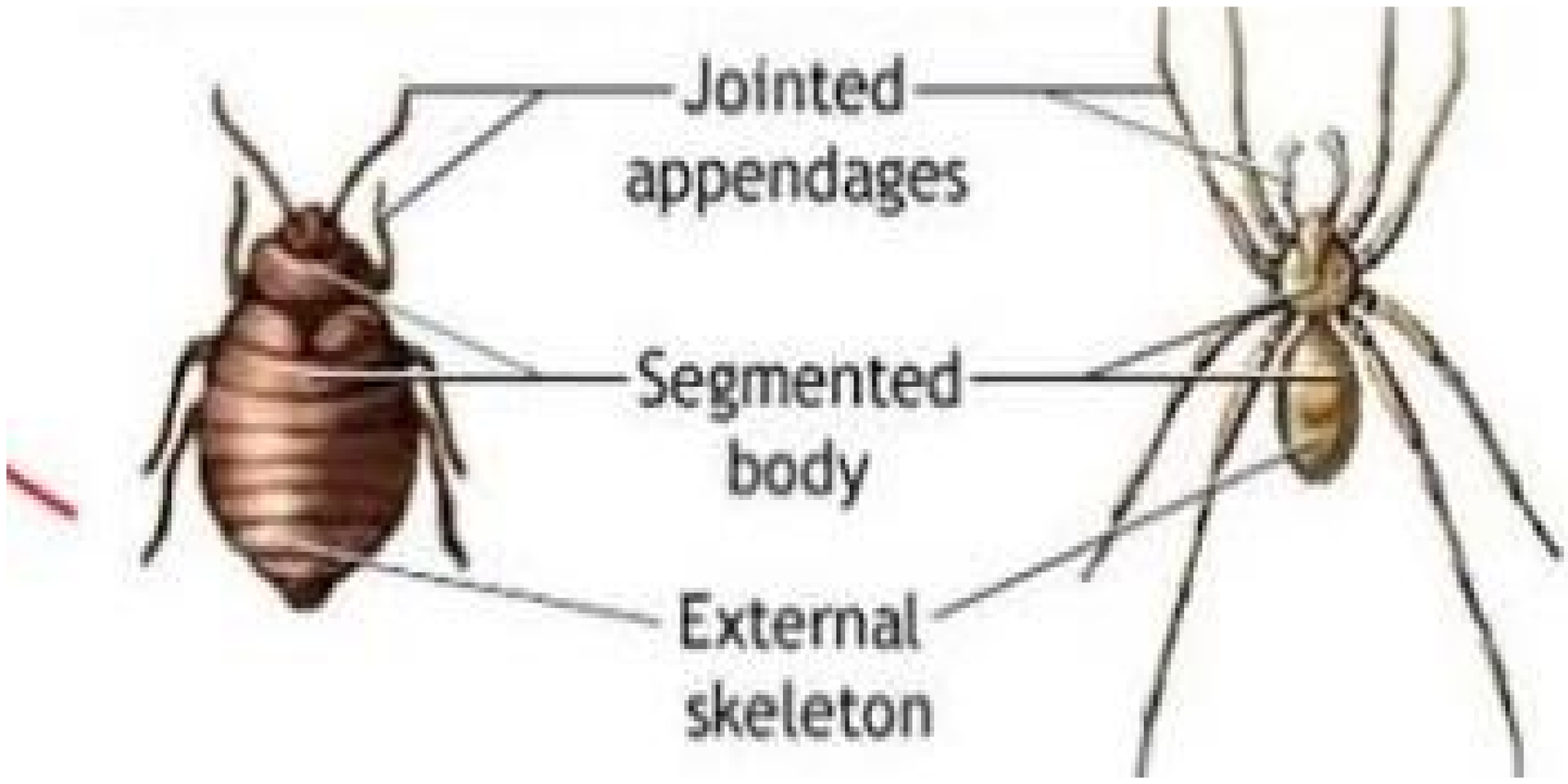
Arthropoda
- *Its members found in the seas, in fresh water, on land or even flying freely*
- *A group with remarkable differences of structure, and so abundant that all the other animals taken together are less than one-sixth as many as the arthropods*
  - *The Well-known members of this group are*
    The lobsters, cray fish and crabs
    Scorpions, spiders, mites, ticks
  - *The centipedes and millipedes Last, and most abundant of all, the insects*

2.Characteristics of Arthropod

Q:.....Characters of arthropoda

One feature will not helps to separate the arthropods from' all other animals But the

characters of an animal of several of these which are described here will enable the observer to determine in each case whether he is examining one of this group.



1.pieces or segments

In arthropods the body is composed of a series of more or less similar pieces or segments Placed one by one, the line of attachment of these to each other being usually somewhat evident on parts of the body at least

e.g. This character is also shown more clearly, in some worms, such as the commonearthworm

2.presence of jointed legs (or appendages):

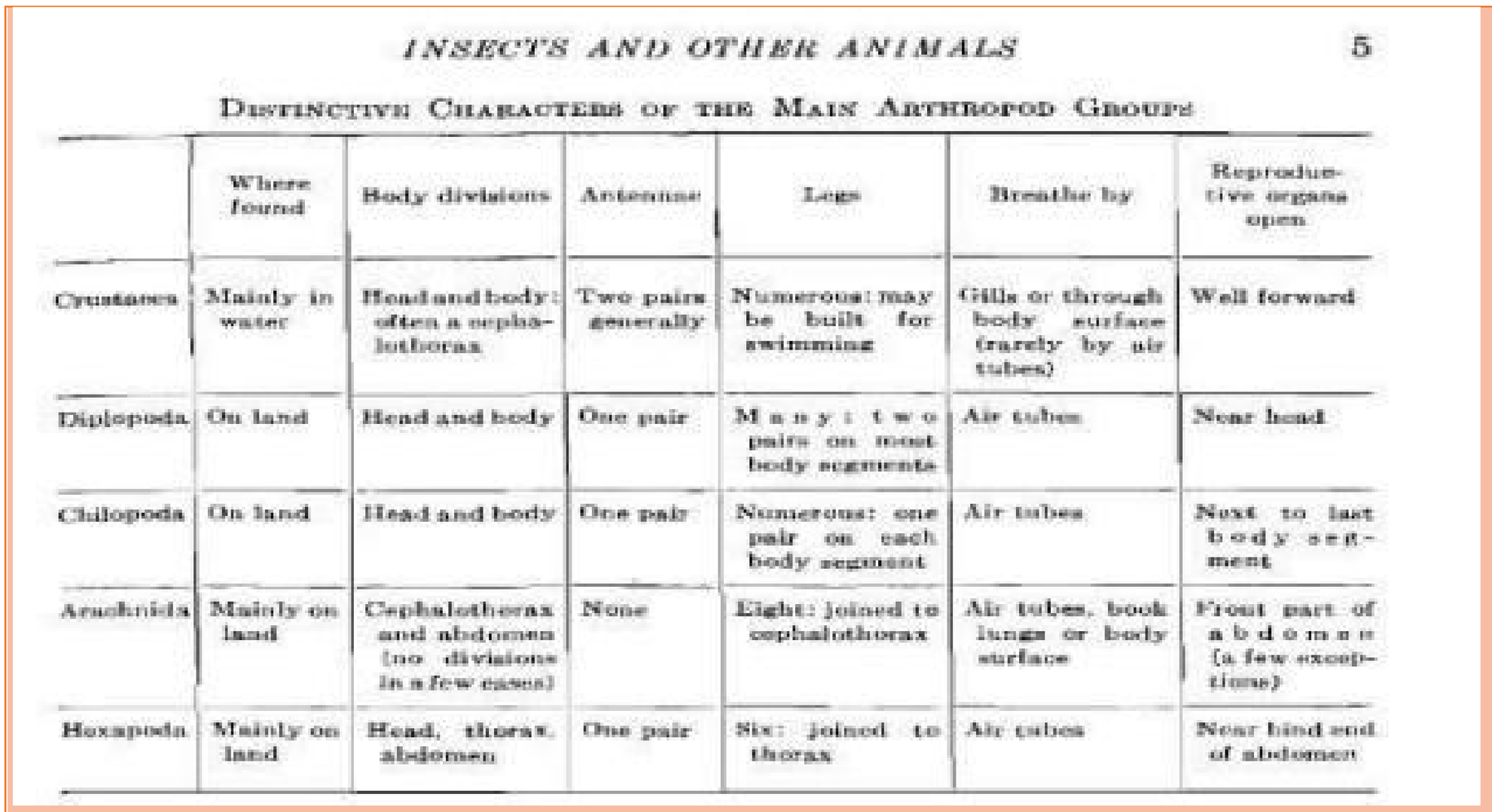
Another character of the arthropods is the presence of jointed legs (or appendages of some kind), as is indicated by the name of the group, and these are not possessed by worms

3.Exoskeleton

The surface of the body is covered by a secretion which hardens on exposure to the air, forming an
outside shell or external skeleton called exoskeleton

There being practically no internal structures supporting the soft organs within this shell except as ingrowths from the outside

Distinctive Characters of Arthropods Groups



INSECTS AND OTHER ANIMALS 5

DISTINCTIVE CHARACTERS OF THE MAIN ARTHROPOD GROUPS

	Where found	Body divisions	Antennæ	Legs	Breathe by	Reproductive organs open
Crustacea	Mainly in water	Head and body: often a cephalothorax	Two pairs generally	Numerous: may be built for swimming	Gills or through body surface (rarely by air tubes)	Well forward
Diplopoda	On land	Head and body	One pair	Many: two pairs on most body segments	Air tubes	Near head
Chilopoda	On land	Head and body	One pair	Numerous: one pair on each body segment	Air tubes	Next to last body segment
Arachnida	Mainly on land	Cephalothorax and abdomen (no divisions in a few cases)	None	Eight: joined to cephalothorax	Air tubes, book lungs or body surface	Front part of abdomen (a few exceptions)
Hexapoda	Mainly on land	Head, thorax, abdomen	One pair	Six: joined to thorax	Air tubes	Near hind end of abdomen

3-Characteristics of Arthropod II

External skeleton:

In the possession of this external skeleton these animals have a seeming resemblance to the shells (Mollusca) But the materials of which the skeleton is composed are quite different Being largely calcium carbonate in the Mollusca

In Arthropods chitin which somewhat resembles horn in its nature, sometimes with calcareous salts·deposited in it, in the Arthropoda .

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Symmetry

The simple members of arthropod have bilaterally symmetrical body, though this condition is concealed somewhat by secondary changes in many of the group . The possession of a bilaterally symmetrical body consisting of a series of segments, an exoskeleton of chitin and the presence of jointed legs are, then, distinctive features of the arthropods .

To separate the various groups of arthropods, other characters must be Used Aside from several five large and important divisions Five large and important divisions there are five large and important divisions which call for recognition

Q : describes the group of arthropods?

Groups of Arthropods 1. The Crustacea, including the lobster, crab, beach flea, sowbug and many Others

2. The Diplopoda or millipedes

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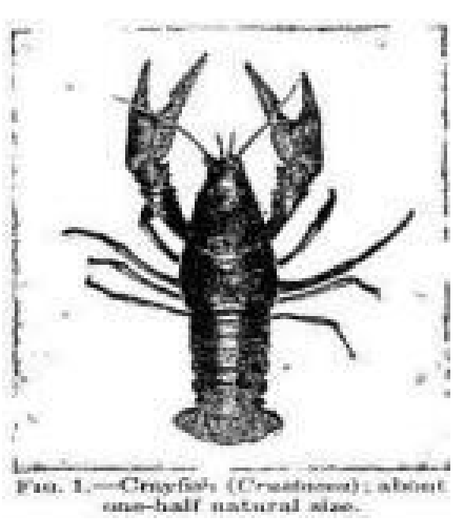
3. *The Chilopoda or centipedes*

4. *The Hexapoda or insects*

5. *The Arachnida, including the scorpions, pseudo-scorpions, spiders, mites, ticks, etc.*

4.Crustacea



Definition:

water Crustacean:

The Crustacea are mainly water-inhabiting animals which breathe either by gills or, in the smaller forms, through the surface of the body ***Land Crustacean:***

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In those cases where its members live on land the gills are still present, though in a somewhat modified condition.

Fig. 2.—Sow-bug; slightly enlarged. A crustacean living on land.

They have numerous pairs of legs and generally two pairs of antennae (jointed "feelers") Q:define cephalothorax??

Often some of the bug; body segments are fused with the head to form a crustacean cephalothorax

5.Diplopoda

Definition:

The Diplopoda are land animals

breathing by air tubes opening on the sides of the body, these tubes carrying the air into all the internal parts of the animal

The head bears a pair of antennae and is followed by a series of segments all practically alike and each, except the first three, with two pairs of legs

The reproductive organs open far forward on the body shape: In most of the more common members of this group the body is quite cylindrical, and when disturbed the animal usually curls up in a sort of close spiral Small diplopods about the diameter of the lead of a pencil and gray in color are often found boring into potatoes and roots in the ground In the fall and are sometimes wrongly called wireworms The Common name, millipede, refers to the large number of legs possessed by these animals

6.Chilopoda

Definition:

The Chilopoda are also land animals Like the diplopods ***Characters:***

they have antennae and breathe by air tubes, and the body segments are practically all alike

The general form, however, is rather flattened Each segment bears only one pair of legs

The reproductive organs open at the posterior end of the body

The front leg on each side is modified to serve as a poison claw

The numerous legs present in these animals have resulted in their receiving the common name, centipede

7.Arachnida

The Arachnida generally have the segments of the body grouped into two sections called

1-The cephalothorax

2-The abdomen

- *No antennae are present and the eight legs are all attached to the cephalothorax*
- *They breathe by air tubes somewhat*
- *similarity with other groups*
- *similar to those of the other groups By sacs containing many thin plates resembling leaves of a book, when these structures take the name of book lungs*
- *In the smallest forms, they directly breath through the body surface*
- *In the mites there is no evident division of the body into sections .*

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- *Though most of the group are land forms, a few are aquatic*

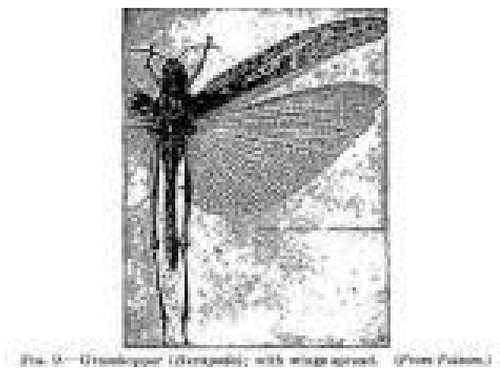
8 .Hexapoda

In the Hexapoda or insects the segments of the body are grouped in three distinct sections:

1. Head

2. Thorax

3. Abdomen



Characters

- *A pair of antennae is present on the head*
- *The six legs are attached to the thorax*
- *The four wings usually present*
- *The animals breathe by air tube*

Insects living under a great diversity of conditions, the group as a whole is

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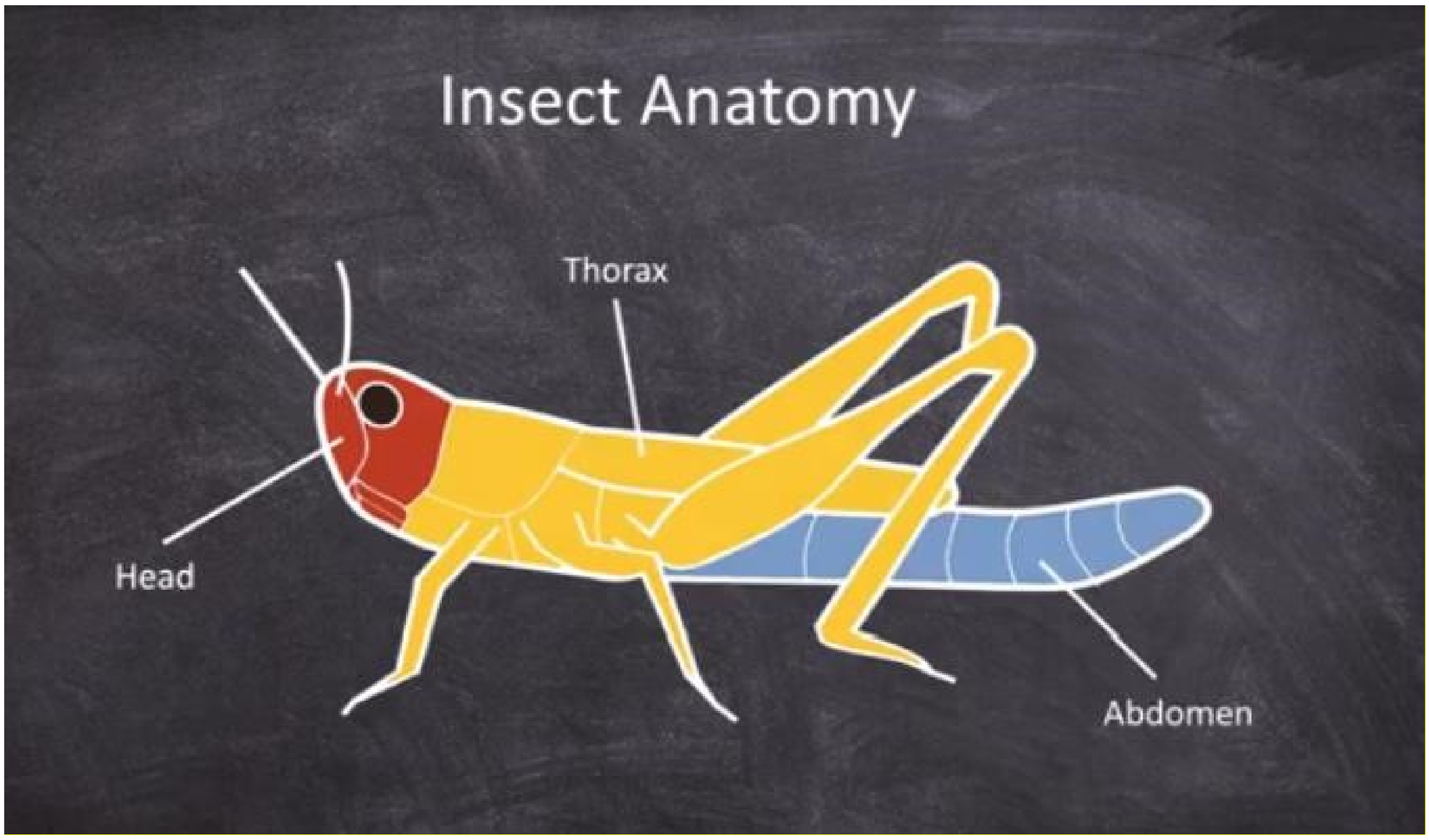
emphatically a terrestrial one though in many cases their early life is spent in water.

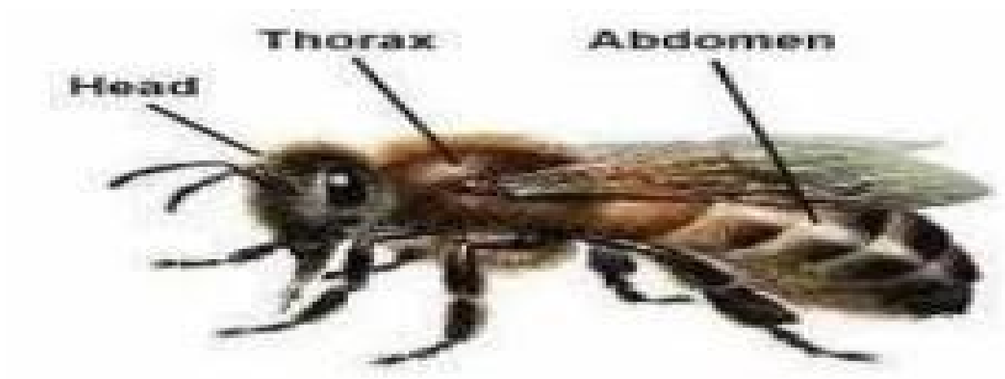
9. General features of insects

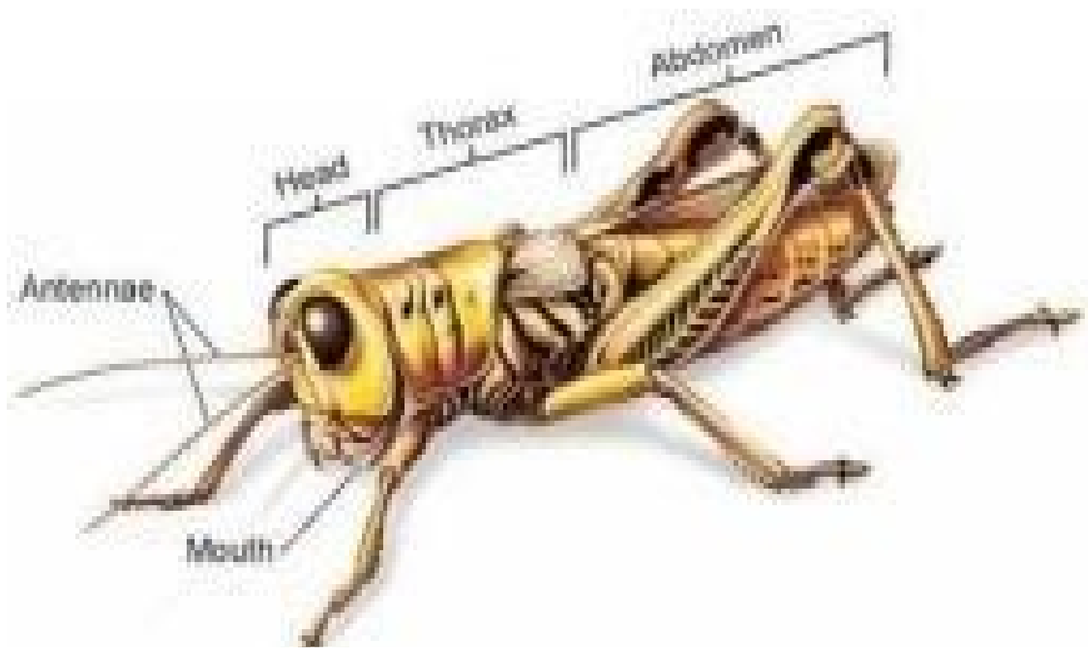
THE INSECT:

ITS EXTERNAL STRUCTURE



Bringing together the facts already stated about insects we find that an adult insect is:

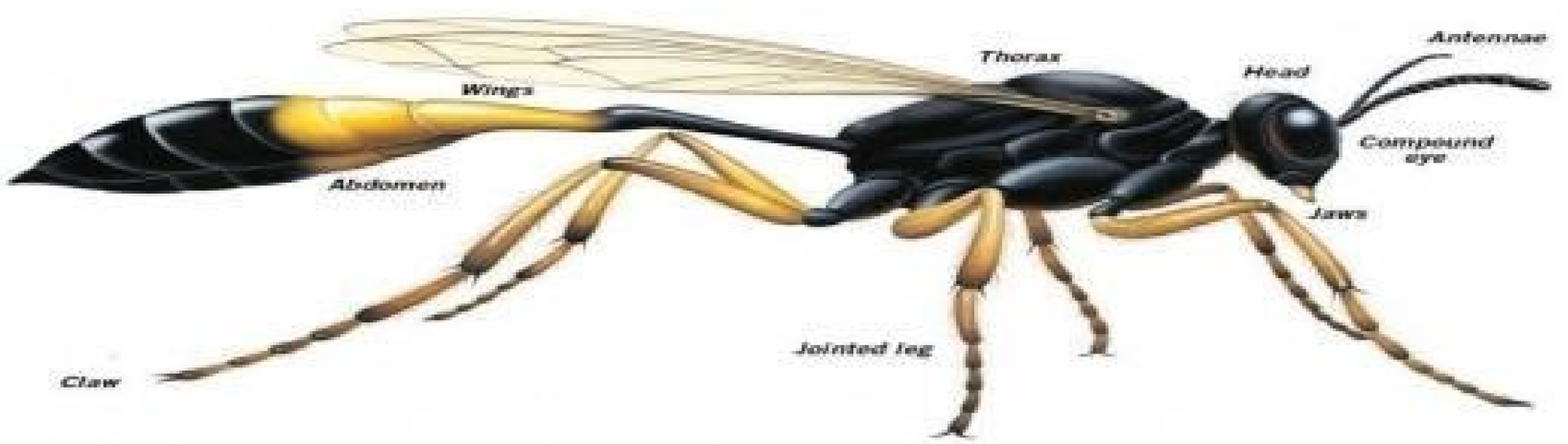




a bilaterally symmetrical animal consisting of a series of segments one behind another;

that these segments are grouped into three regions, The head in front Followed in order by the thorax The abdomen that the animal is covered by a skeleton, shell-like in that it is on the outside of the body, but horny in its nature

- three pairs of jointed legs attached to the thorax



- a pair of antennae

- mouth parts

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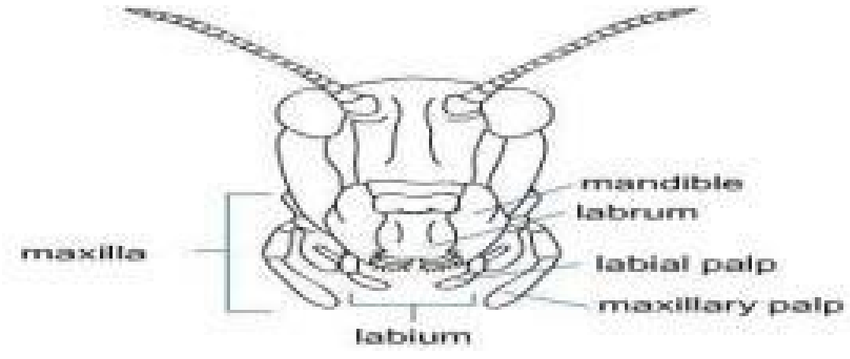
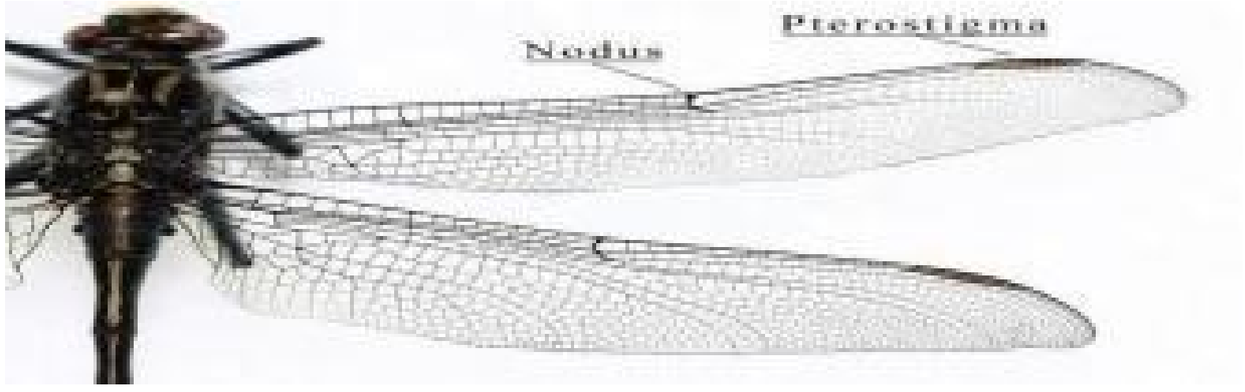
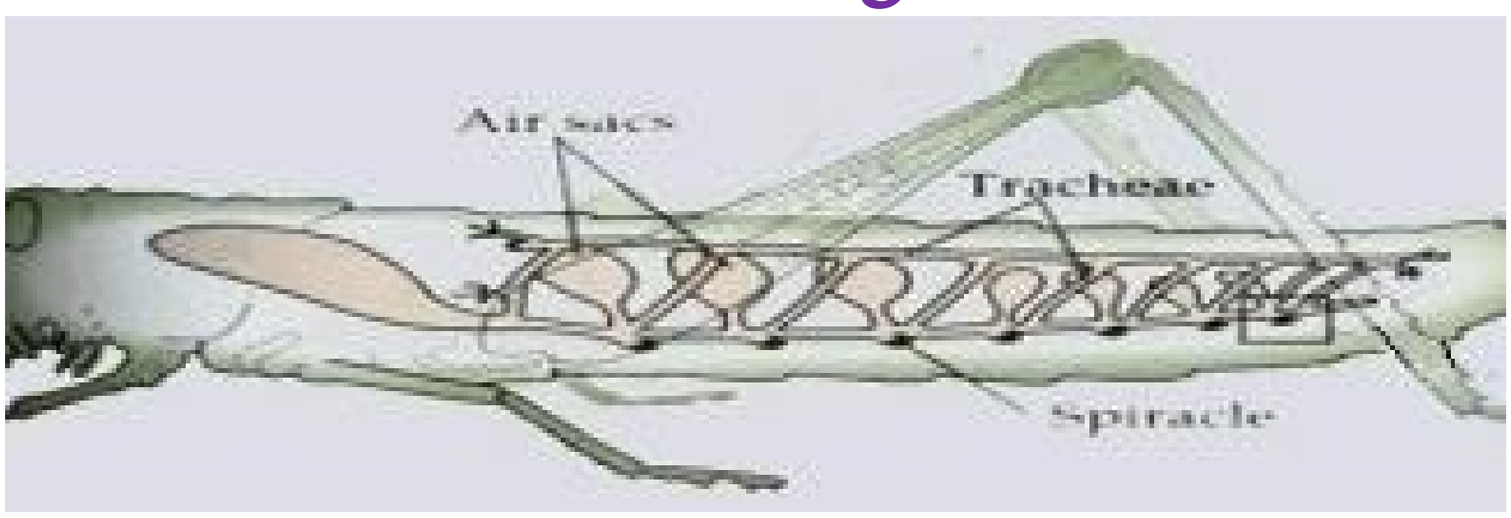


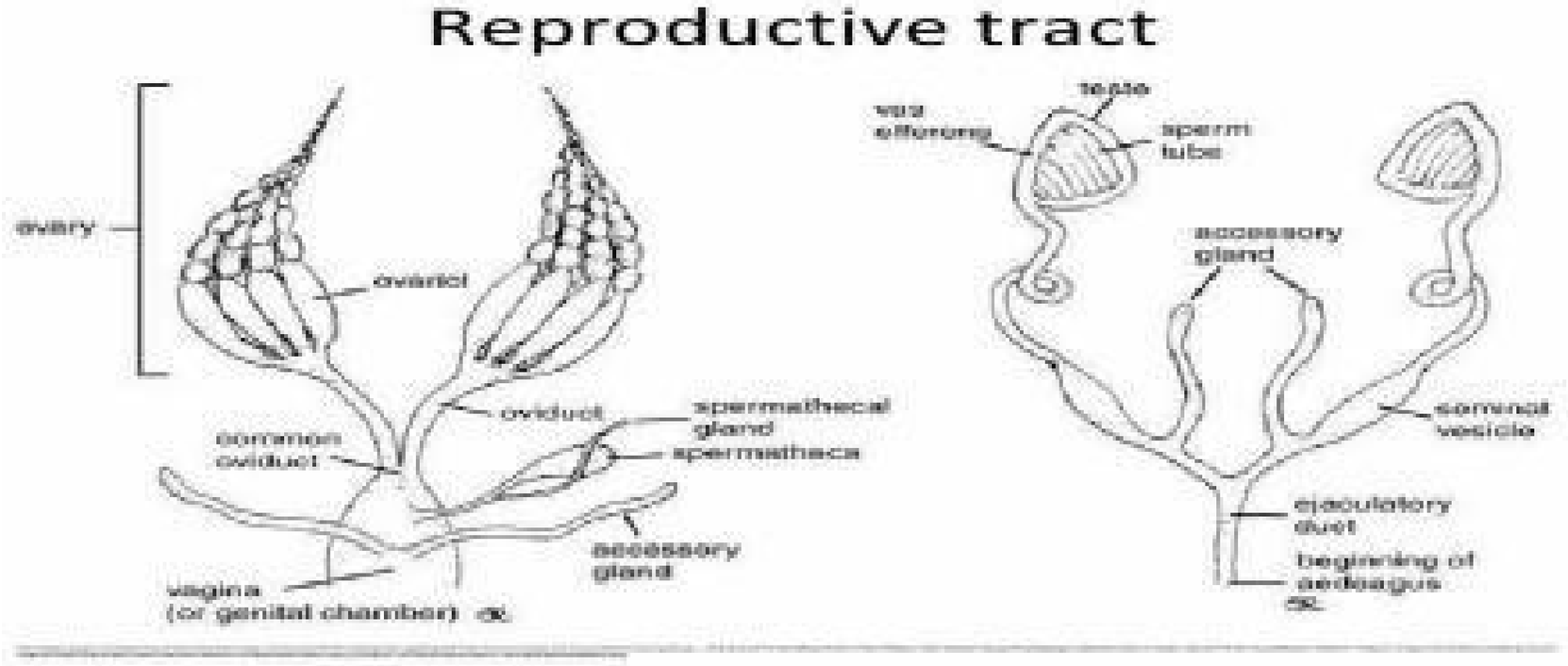
- *and usually two pairs of wings are present*



that it breathes through air tubes



that the reproductive organs open near the hinder end of the body.



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Q: Characteristics of insects??

- Bilaterally symmetrical.
- Consisting of a series of segments.

(a) head

(b) thorax

(c) abdomen.

- The animal is covered by a Shell like skeleton present outside the body, horny in nature.
- Three pairs of jointed legs attached to the thorax.
- pair of antenna.
- Mouth parts. two pair of wings.
- Breathing through Air tubes.
- Reproductive organs present near the hinder end of the body

11. General segmentation of insects

Segments:

Q: how many segments are present in egg??

in egg

In the body of the adult insect there are apparently from

Head

7 to 15 segments 1 in the head,

Thorax

3 in the thorax

Abdomen

,3 to 11 in the abdomen In the egg, however, the

Q:how many segments are present in embryo??

embryo

shows the presence of 6 segments in the head , 3 in the thorax and 11 or 12 in the abdomen Indicating that the original total number was 21

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Q:what happen to segments when embryo changes to adult? adult

This change to the adult has been brought about in the head by an extreme amount of fusion and condensation and in the abdomen partly by fusion Partly by a sort of telescoping or gradual shifting of one segment within another until it has been partly or entirely concealed

12-13 cuticle or external skeleton

Q: what is cuticula??

The skeleton of the body, or cuticula

- *It supports and protects the soft, living tissues within it*
- *It rests on the outer surface of a layer*
- *It seems to be formed in part from a fluid poured out from the hypodermis and in part by a transformation of a portion of the hypodermis cells themselves*
- *Q: how exoskeleton gets harder??*
- *The cuticula hardens quickly after its production and consists quite largely, at*

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least, of a nitrogenous substance called chitin

- It varies in thickness and is flexible where movement is needed, but thicker, more rigid and usually darker colored elsewhere
- The hypodermis along certain lines forms folds which project inward, carrying its cuticular secretion along with it Where these infoldings occur at places that need to be movable the cuticula remains flexible Thus forming the movable joints; along the other infoldings it hardens (becomes sclerotized) and
- Q: define sutures??becomes rigid and such places show on the surface only as slight grooves or scratches called sutures
- Q:what are plates or sclerites?
- These sutures and the joint lines divide the cuticula into areas called plates or sclerites
- The sutures and joint lines are sufficiently regular in position in most insects to make them convenient landmarks in descriptions
- Q:name the scleritis found in back,on each side and below the segments???

- *The sclerites on the back of each segment are usually together termed the notum or tergum; those on each side, the pleuron; and those below, the sternum*
- *Q:give the amount of sclerites in head,thorax and abdominal regions???*
- *In the head the sutures are few in number, so only a few plates or sclerites are generally in evidence*
- *In the thorax they are more numerous, and*
- *in the abdomen often only a dorsal and a ventral sclerite for each segment are found Occasionally weakly sclerotized areas are quite large (queen white ant) and elastic.*

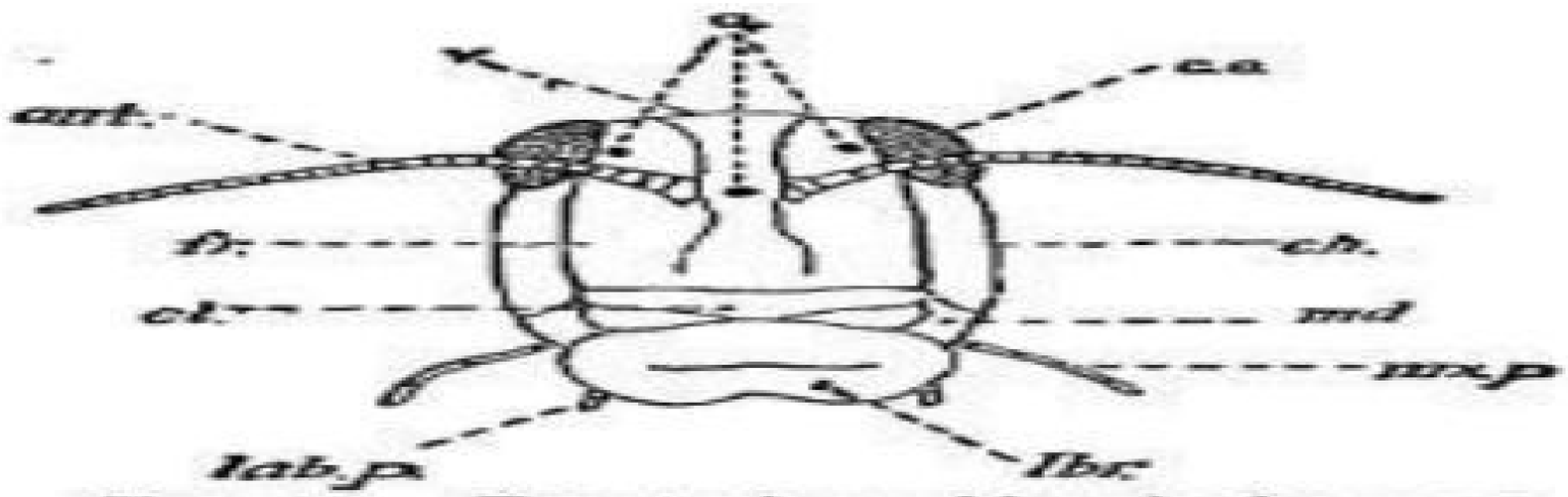

FIG. 11.—Front view of head of a grasshopper showing a hypognathous head: *ant.*, antenna; *c. e.*, compound eye; *ch.*, cheek; *cl.*, clypeus; *fr.*, frons; *lab. p.*, labial palpus; *lbr.*, labrum; *md.*, mandible; *mx. p.*, maxillary palpus; *o.*, ocelli; *v.*, vertex.

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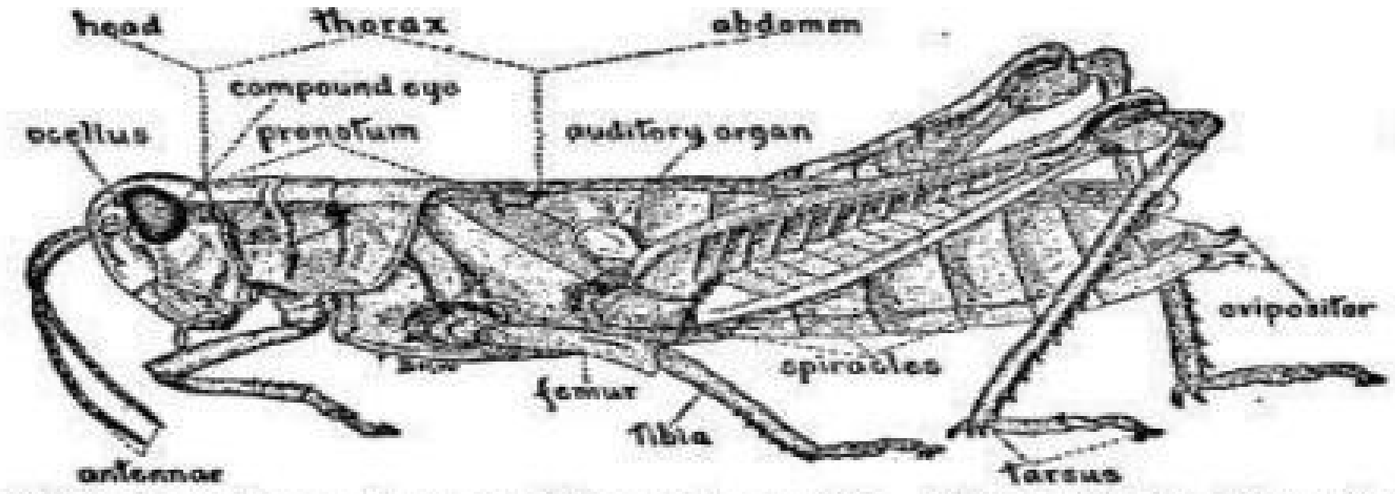
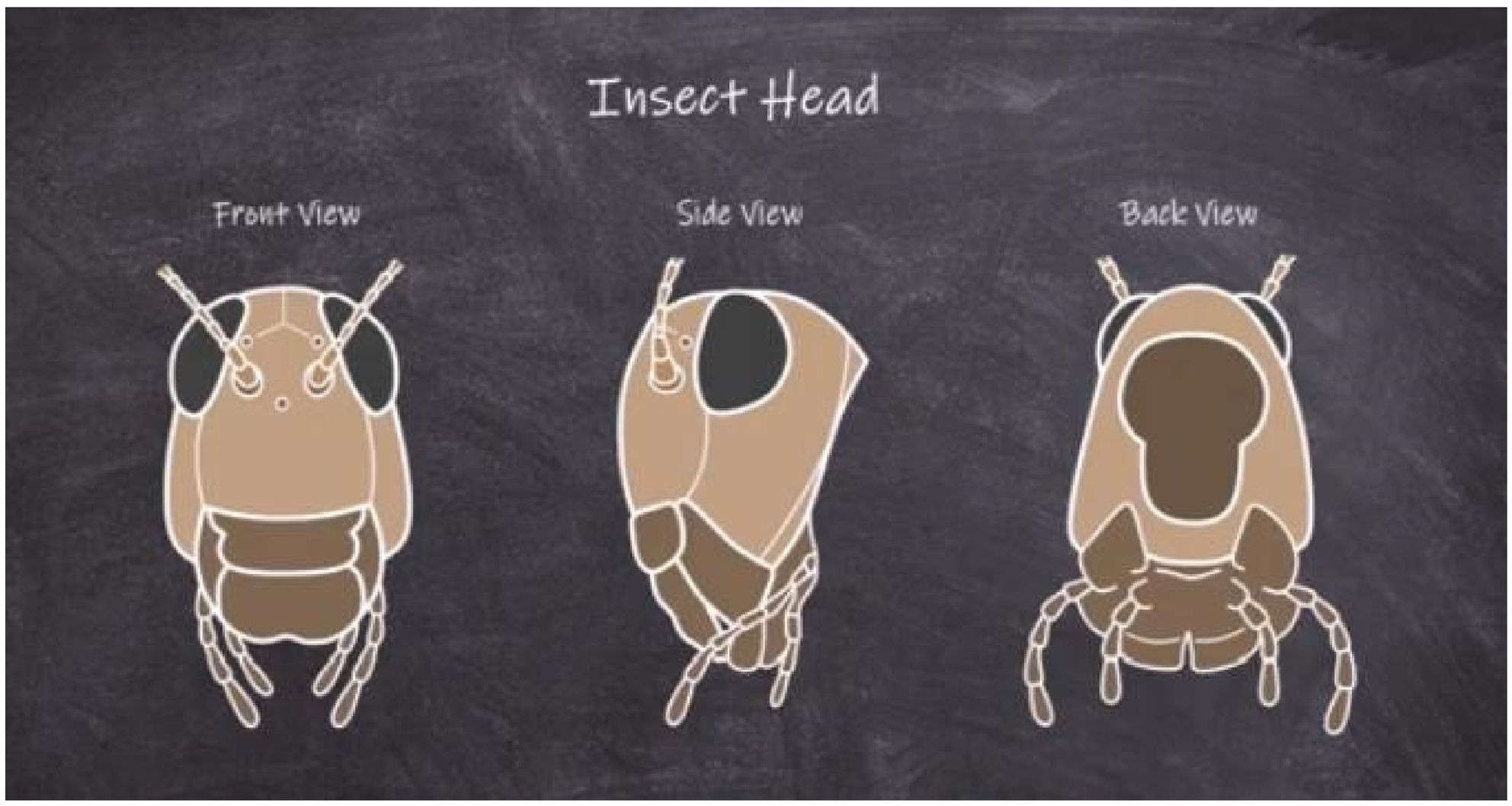


10.—Side view of grasshopper with parts named. *(From Walden, Conn. Geol. & Nat.*

14. Structure of Head

Q: describe the different heads types in insects?/define head of insects and it's types??

The heads of different insects vary much in form and in the location of the mouth



In some cases this is on the underside; in others it is practically on the front.

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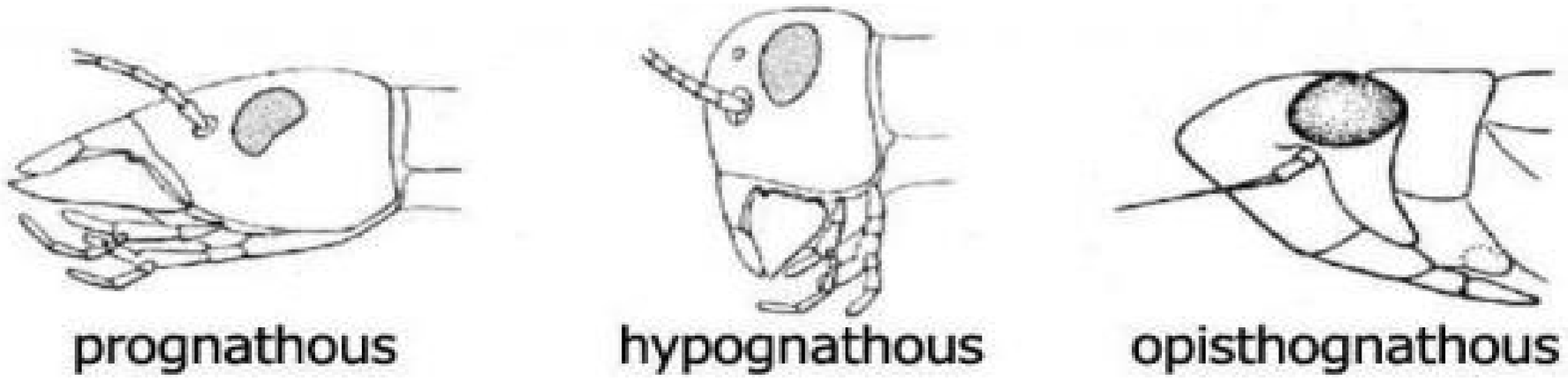
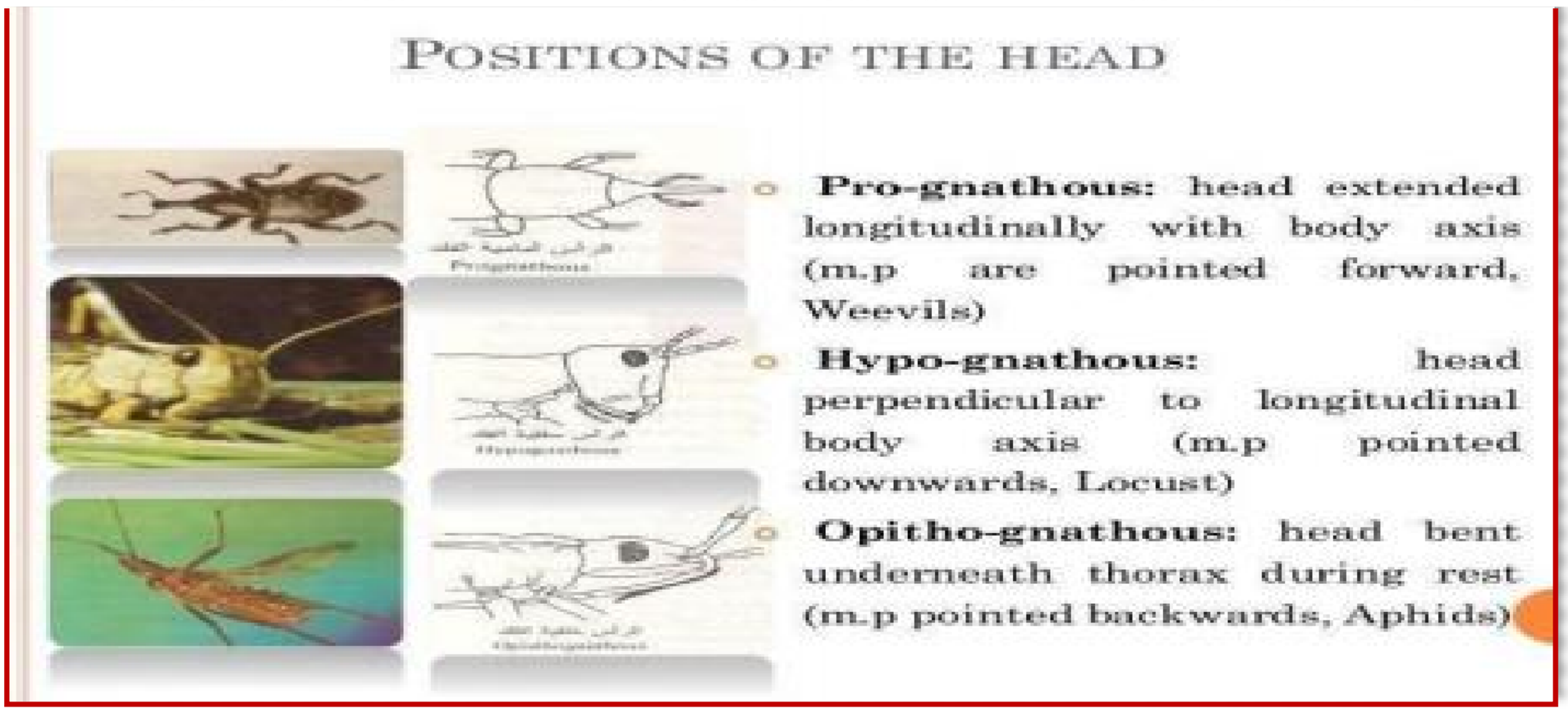
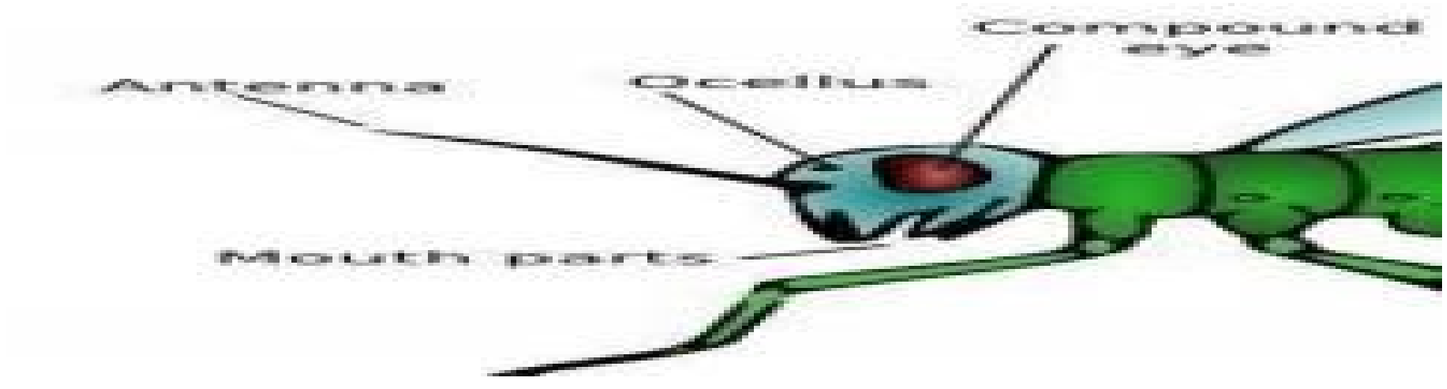




Heads with the mouth beneath are called hypognathous those with it in front are prognathous. Structures found on the head are

★ *A pair of antennae* ★ *Two compound eyes (ocelli)*

★ *The mouth parts*



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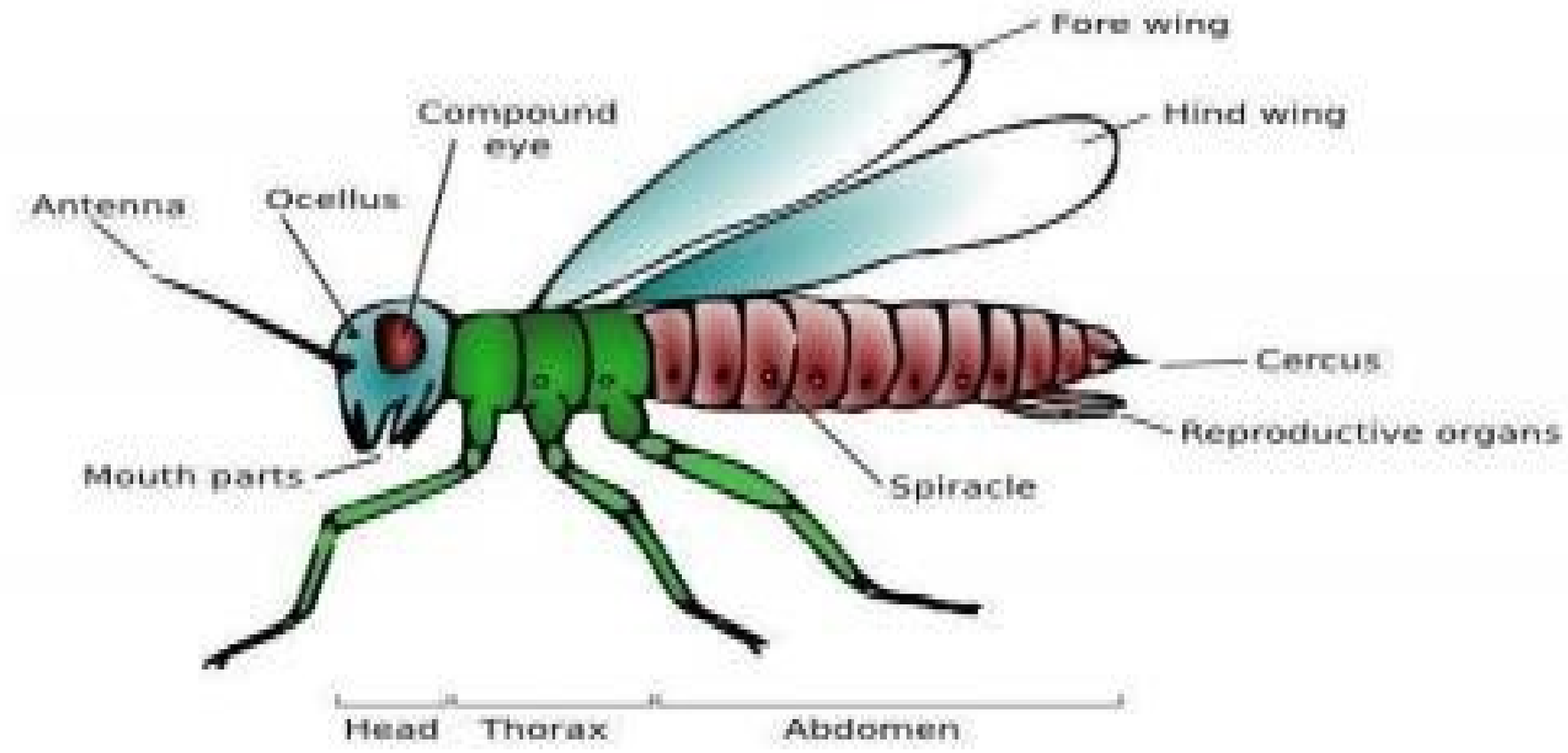
On the thorax are the wings and legs



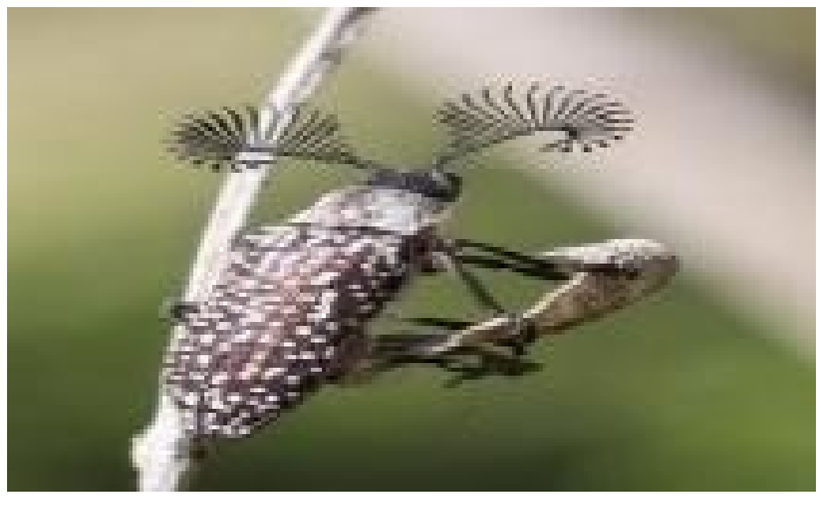
; on the abdomen

◆ *are various organs such as* ◆ *the ovipositor,* ◆ *sting* ◆ *cerci*

◆ *and styli, present in some cases, absent in others.*

15: Antenna

Antennae are nearly always present



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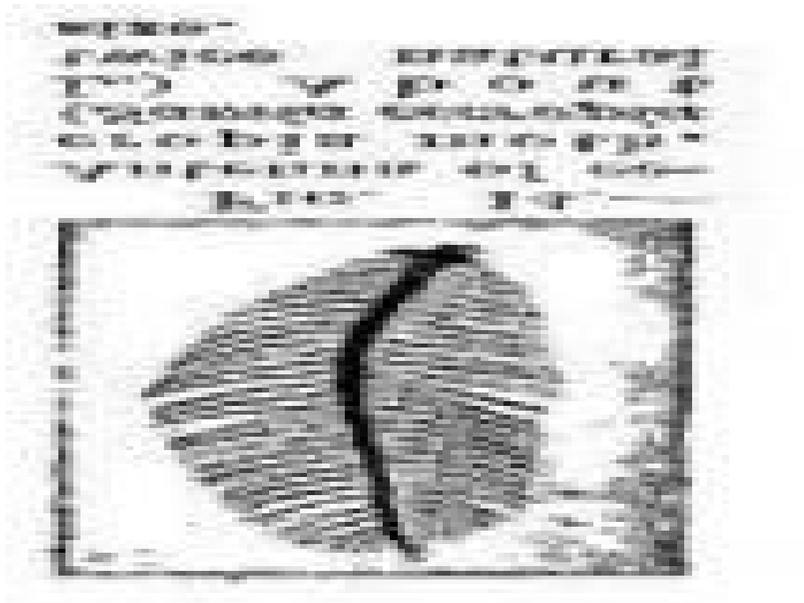

They are usually slender, jointed Therefore more or less flexible organs, varying greatly in the number
of segments composing them
They are
sometimes very short
sometimes long;
often thread-like;
sometimes enlarged near the tip;
in many cases with fine branches either on one or both sides, so that they resemble feathers or plumes; rarely they

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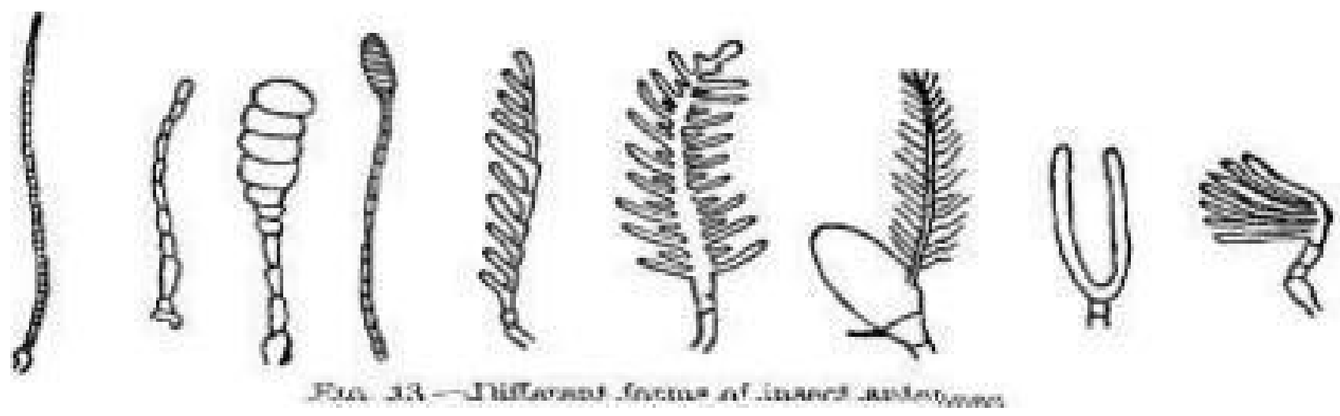
fork; in fact they are of many forms
Sense organs are present on them
for the sense of touch, and probably also
for smell and hearing, at least in Some cases



16: Eyes

Q:write a note on two kinds of eyes 👀 in insects?

The eyes are of twokinds

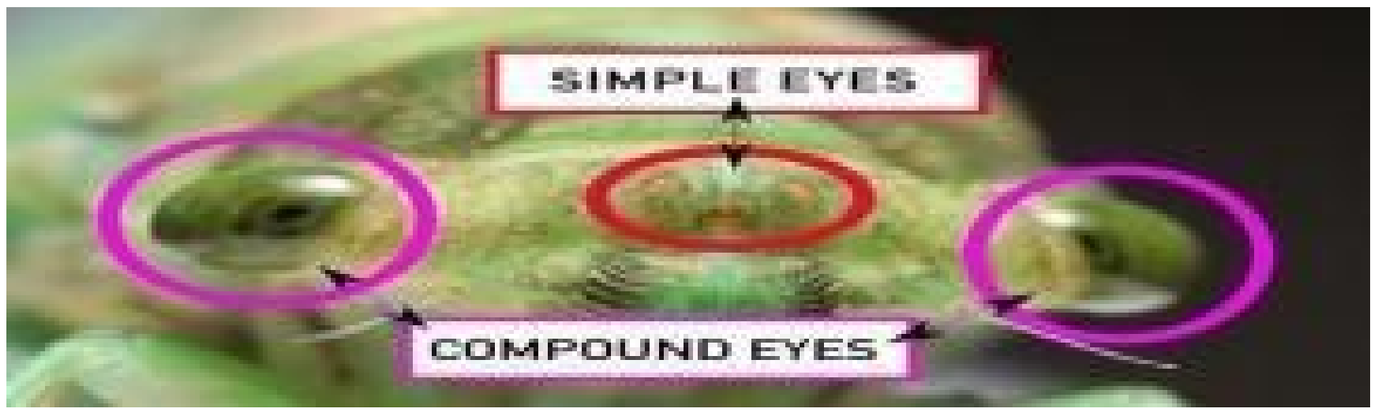
There is a pair of compound eyes, each of which is a group of similar structures which usually are like tall, slender pyramids in form

Q: Define facets/ faceted eyes??

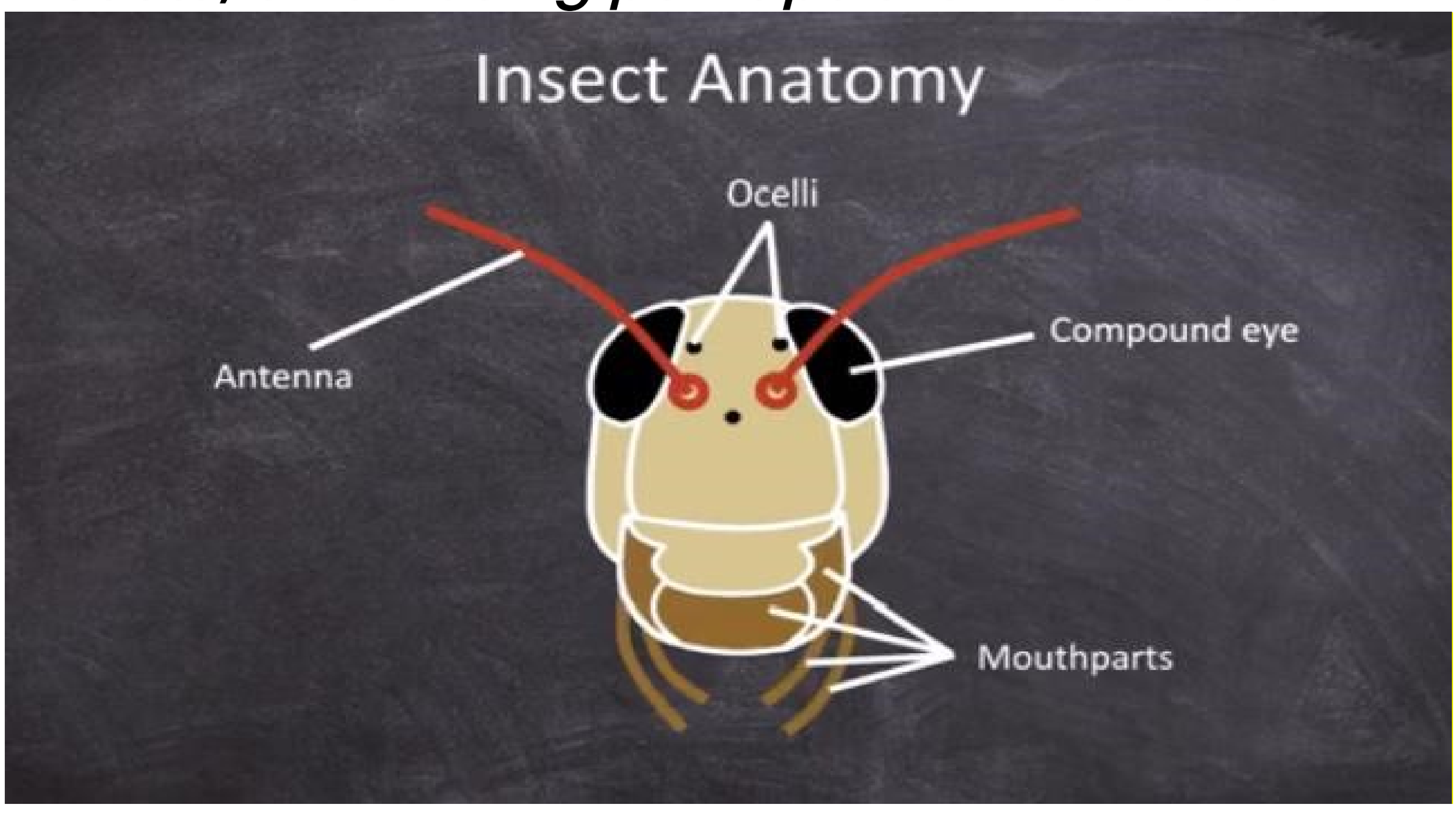
nly the bases of these pyramids Show on the surface, the remainder being within the head The bases, closely pressed together, are usually more or less hexagonal, and their outlines can often be easily seen with a magnifying glass

They are called facets, and the eyes themselves are sometimes termed the faceted eyes **Q: Define ocelii??**

The other kind of eyes, called Ocelli, may be absent or, if present, may vary in number in different



insects, three being perhaps the most usual.



Each, as seen from "the surface, is a nearly circular, convex spot about the size of one of the facets of a compound eye It may be larger than this but is never equal to an entire compound eye in size

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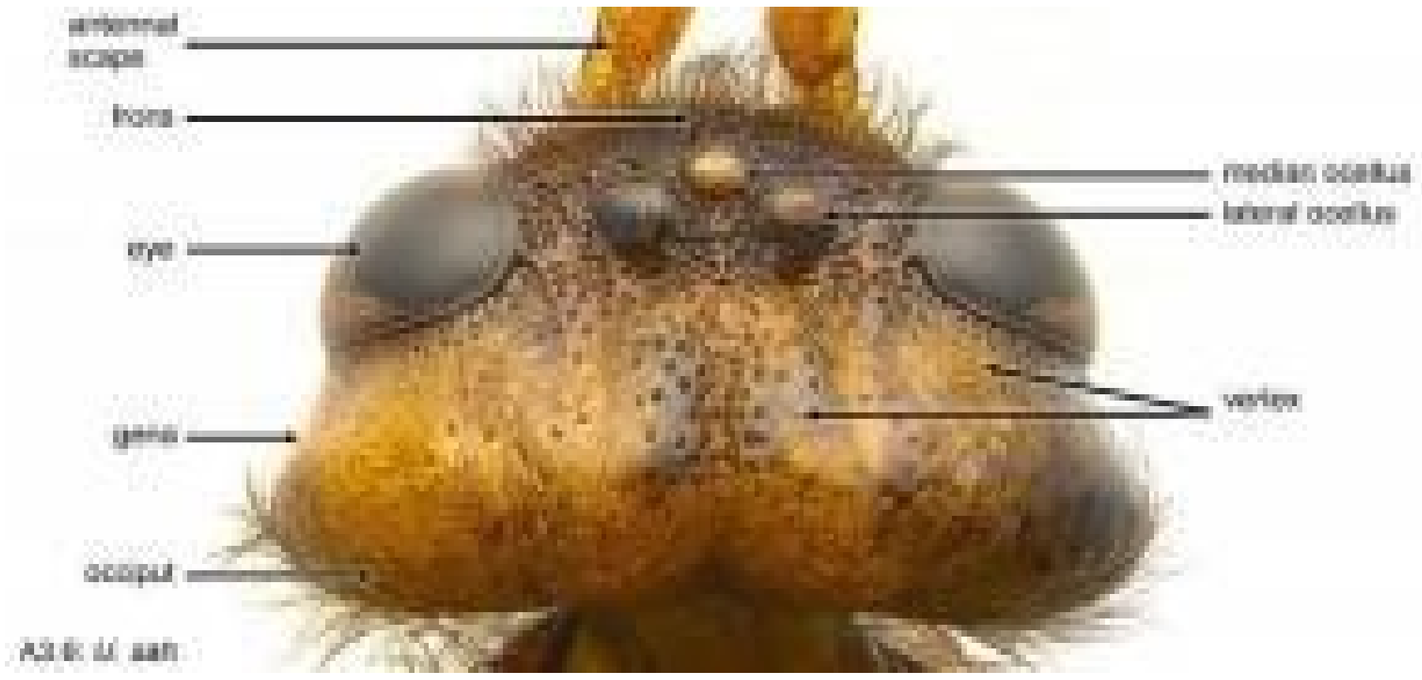

The cuticula of the surface of the body is transparent where

function of cuticula

★ *it covers the surface of an eye,*

★ *permitting access of light to the sensory structures within;*

★ *elsewhere it is usually pigmented and rather opaque*

17-18 -19: Mouth parts

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Q: how many Mouth parts in insets??

1. Labrum

In front of (in hypognathous heads) or above the mouth opening (prognathous heads) is the front lip or labrum It is a thin flap, hinged to the skeleton of the head, and moves forward and backward

2.labium

Behind the maxillae and closing the mouth opening behind, is the hinder lip or labium

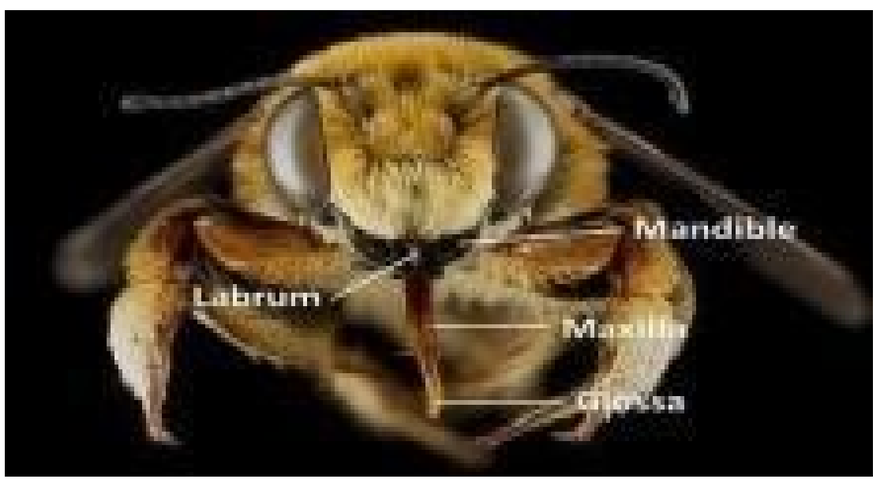
3.Mendibles:

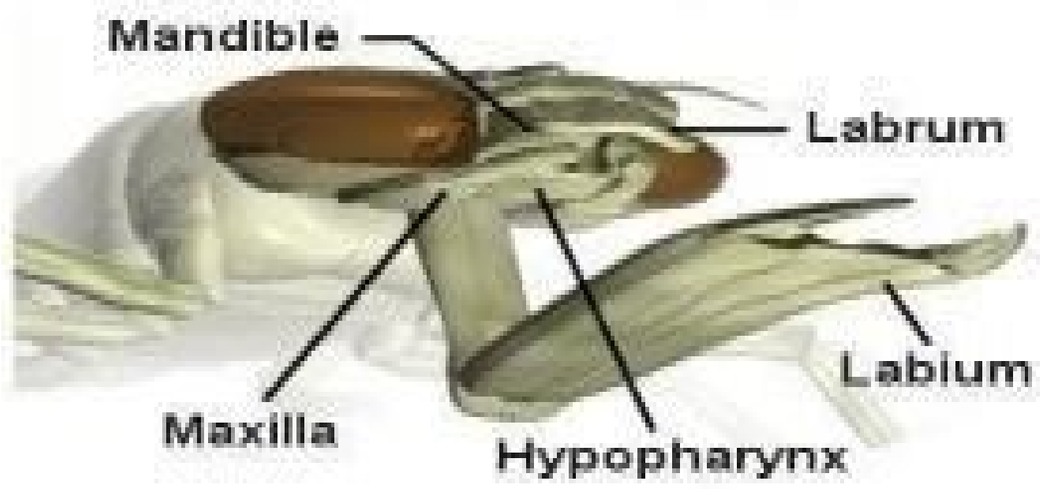
At the sides of the mouth opening, immediately behind the labrum, is a pair of jaws, the mandibles

4.Maxilla:

Immediately behind each mandible at the side of the mouth is a second appendage, the maxilla

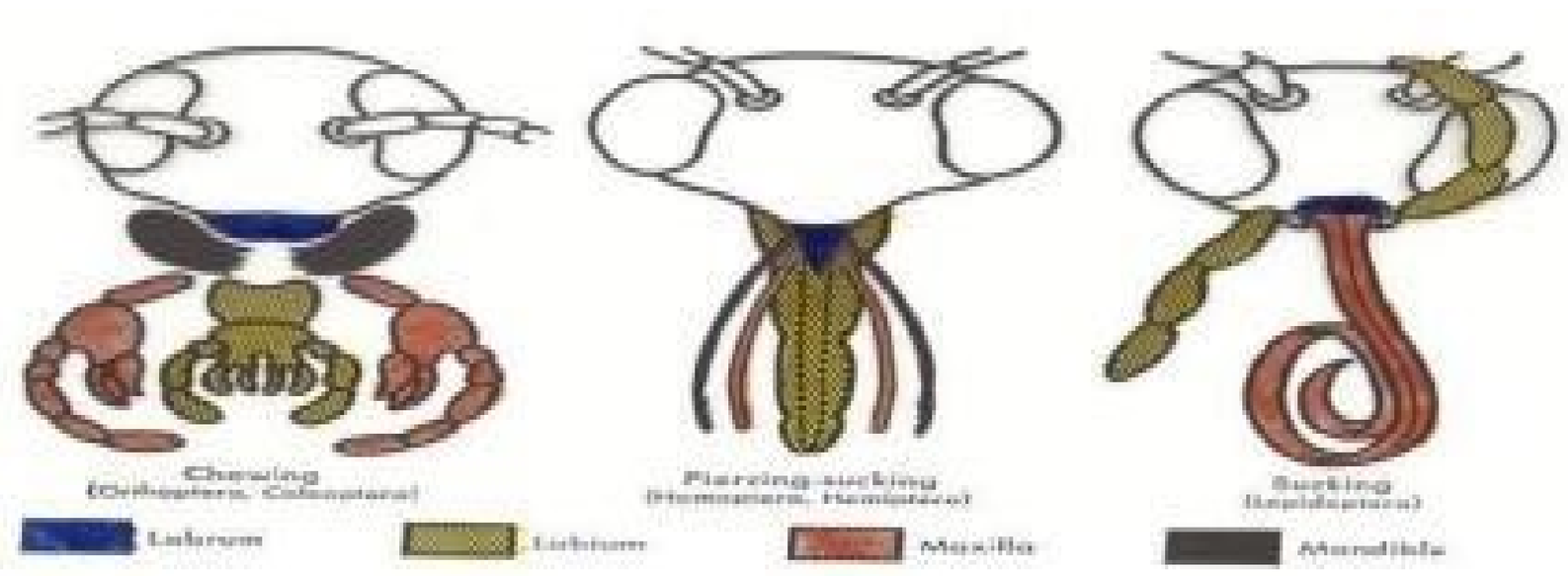
The mouth parts of insects vary extremely in their structure Apparently the original mouth parts were
for biting and chewing, and this type is very common





In some groups, however, they have been transformed into a sucking apparatus

Biting mouth parts, being the more primitive and

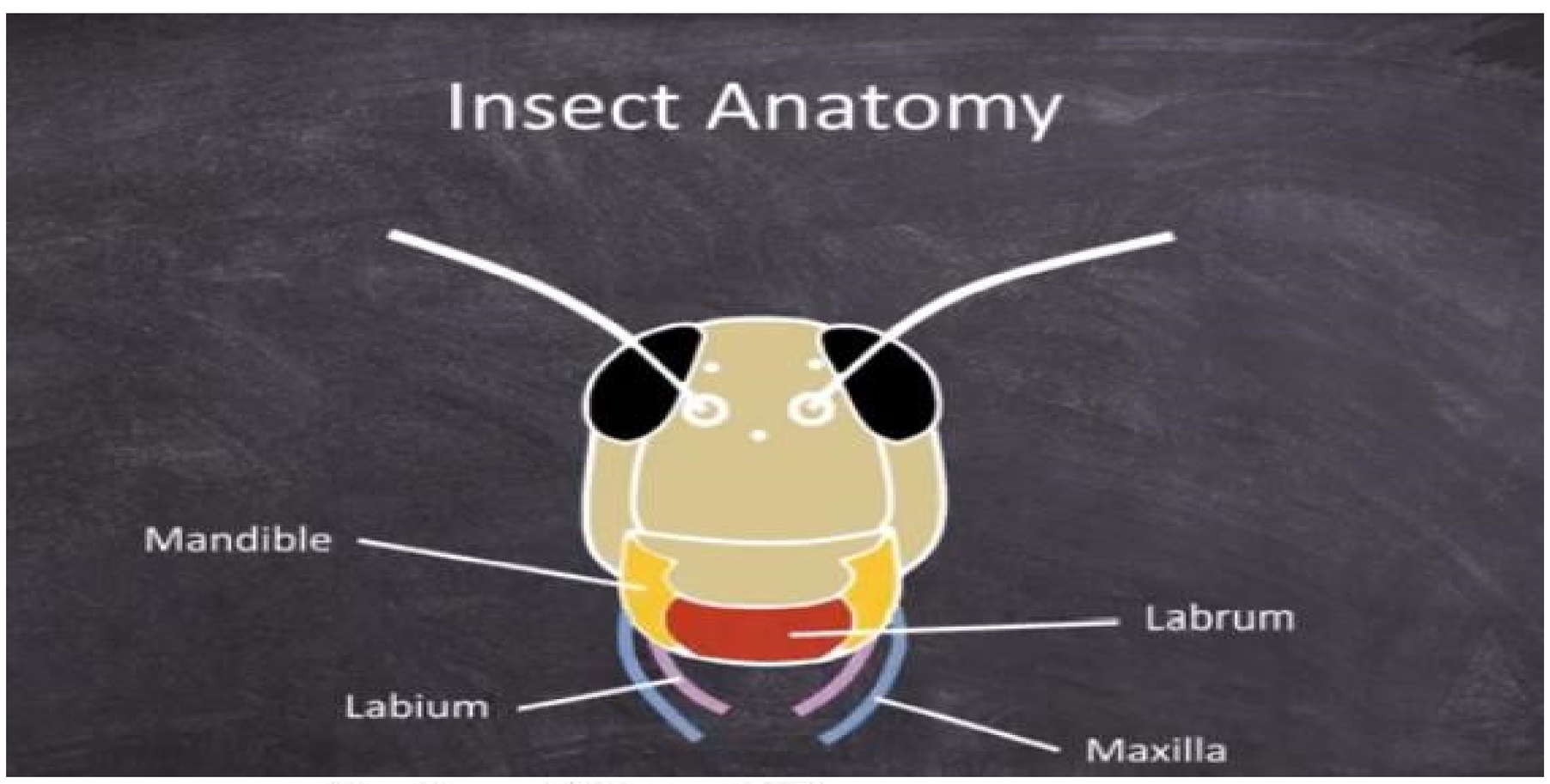


simple, are described here

While sucking mouth parts having been differently transformed in different groups will be taken up in connection with those groups

In front of (in hypognathous heads) or above the mouth opening (prognathous heads) is the front lip or labrum It is a thin flap, hinged to the skeleton of the head, and

moves forward and backward Attached on the outer side of each maxilla, not far from where the latter articulates with the head, is a sort of tiny antenna like structure consisting of from one to six (usually five) segments, which is called the maxillary palpus Behind the maxillae and closing the mouth opening behind, is the hinder lip or labium



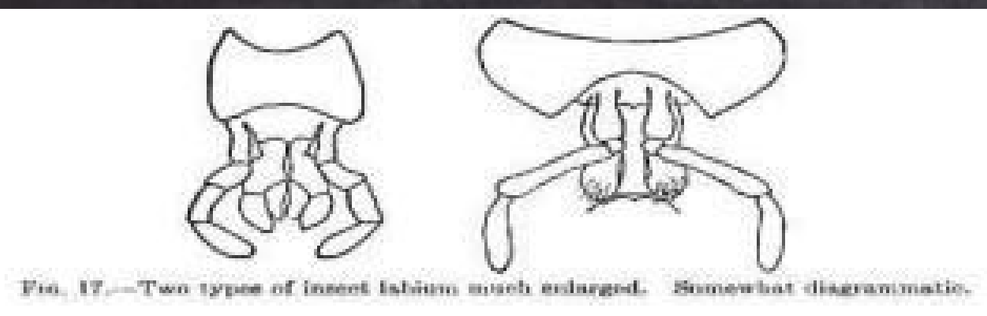
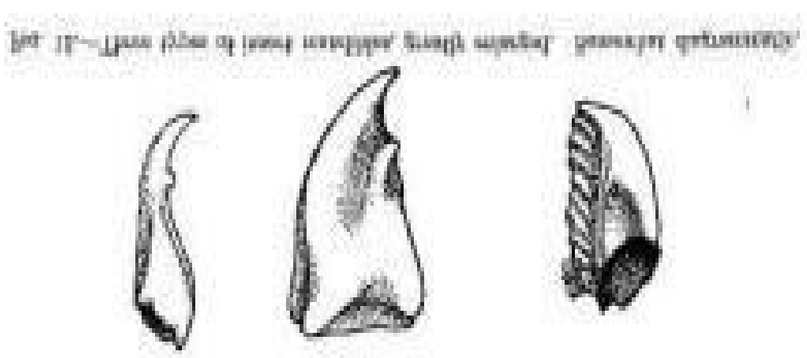
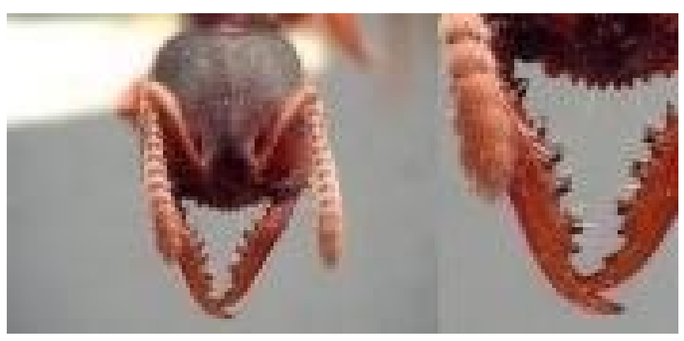


FIG. 17.—Two types of insect labium much enlarged. Somewhat diagrammatic.

It is often more or less divided by a central notch at the middle of its free edge Its inner surface, forming the roof of the mouth, is often called the epipharynx At the sides of

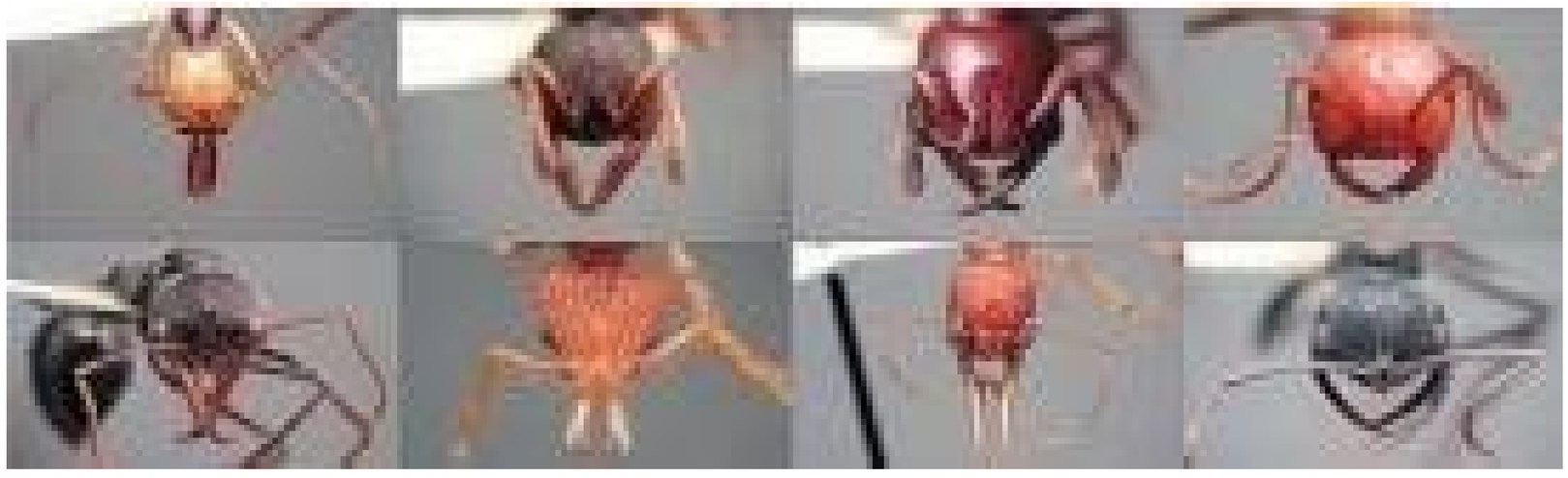
the mouth opening, immediately behind the labrum, is a pair of jaws, the mandibles These differ greatly in form in different insects





They are often stout, heavy structures with crushing faces bearing blunt projections or teeth; sometimes they are long, curved and rather slender In general their form is adapted to the feeding habits of the insect Immediately behind each mandible at the side of the mouth is a second appendage, the maxilla This differs markedly from the mandible) being much weaker and composed of a number of pieces

The tips and outer internal margins of the maxillae usually bear numerous spines or hairs But this condition varies according to the nature of



the food of the insect

This was evidently once a pair of jaws somewhat similar to the maxillae, but with no mouth cavity between to separate them their inner edges have grown together to varying degrees in different

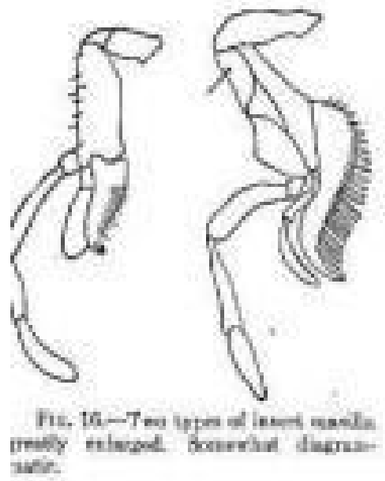
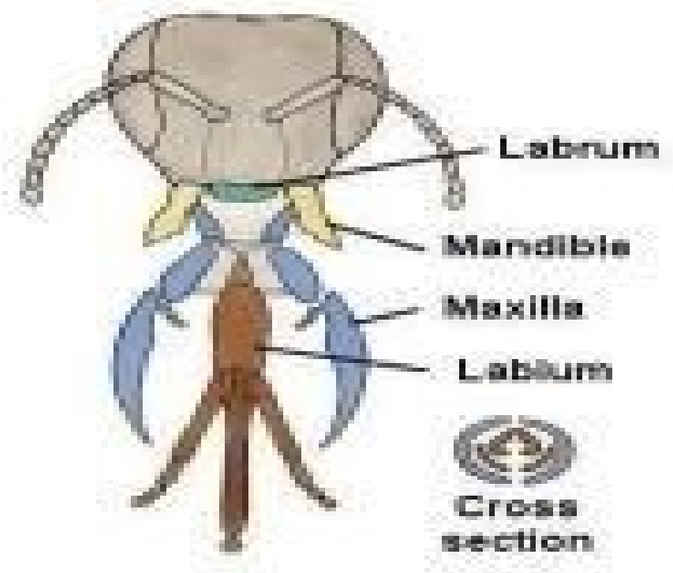


FIG. 16.—Two types of insect maxilla greatly enlarged. Somewhat diagrammatic.



insects

In some, only one or two of the pieces nearest the head have fused In others, fusion all the way to the tip has been accomplished; and all intermediate stages also occur Thus producing a structure which now moves forward and backward

like the front lip, but which may be complete, or partly or almost entirely cleft in the middle line The function of the maxillae appears to be

★ to hold and retain the food in the mouth while

★ it is being worked upon by the mandibles, and also to aid these in breaking it up

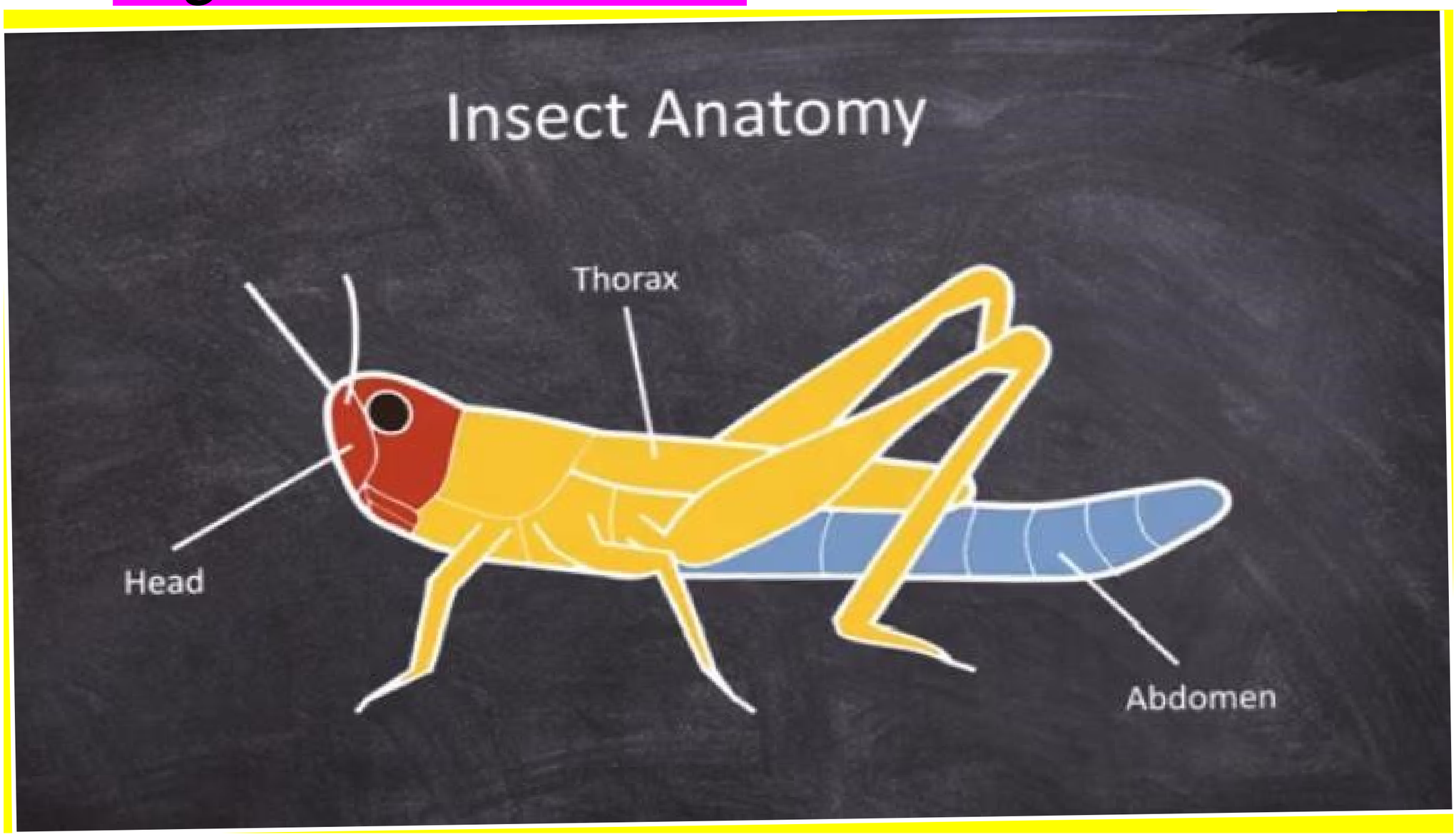
★ The presence of sense organs on the maxillary palpi suggests that these are possibly concerned with the sense of smell Both mandibles and maxillae move sideways Like the maxilla the labium has li, pal pus on each side arising from near its base, and composed of three (rarely four) segments The function of these labial palpi appears to be similar to that of the maxillary palpi

Q Define :Hypopharynx ??

Near the base of the labium on its inner or mouth side there is frequently a fleshy swelling more or less covered by bristles or hairs, which is called the hypopharynx, lingua or tongue It varies greatly in size and form

20-Structure of Thorax

define thorax and also explain its Structure??/what do you know about the segments of thorax??



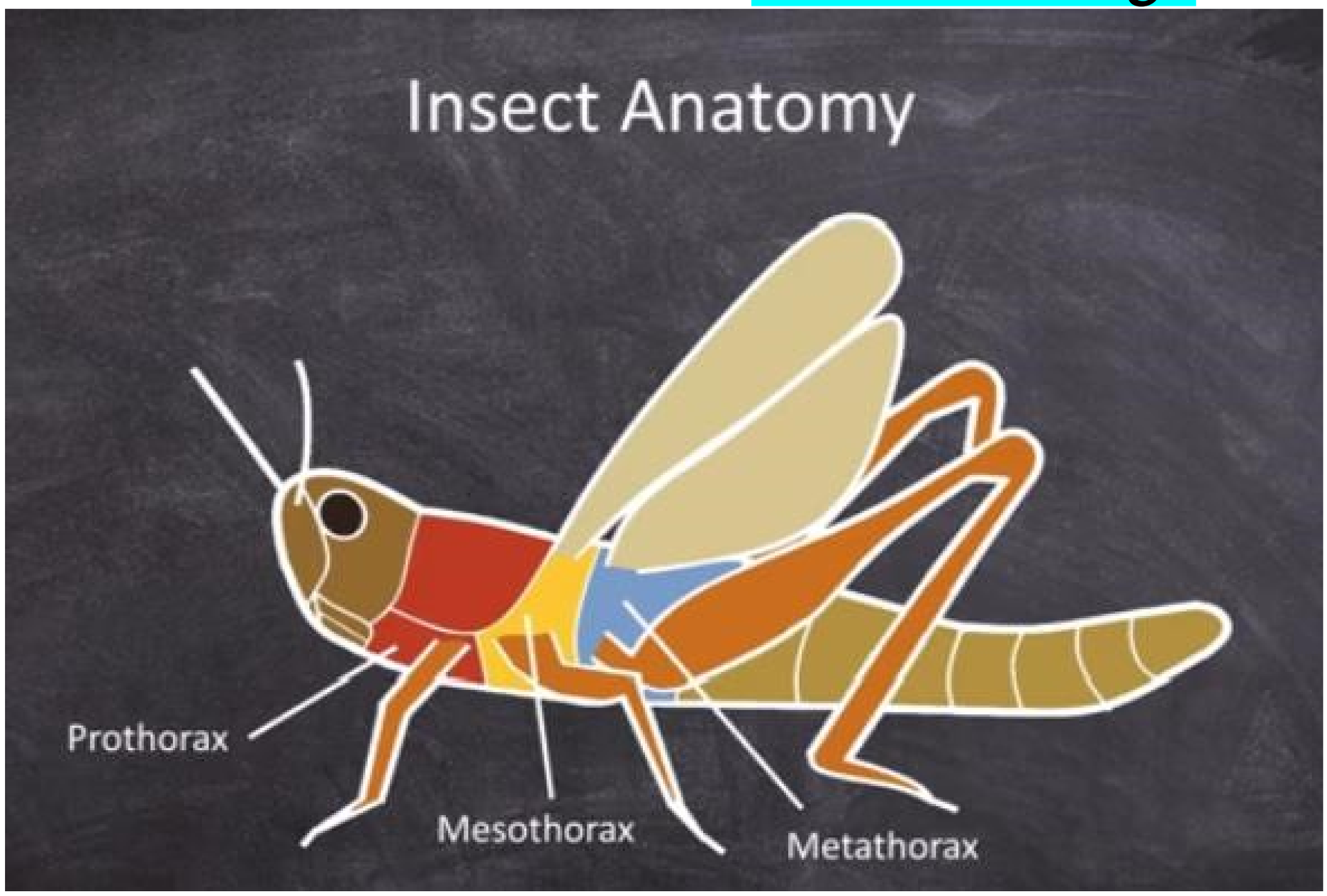
The thorax has its three segments usually quite clearly marked

1-Prothorax

2-Mesothorax

3-Metathorax

Each segment.contains
a pair of legs, but the prothorax, or first of the three behind the head,Bears no wings



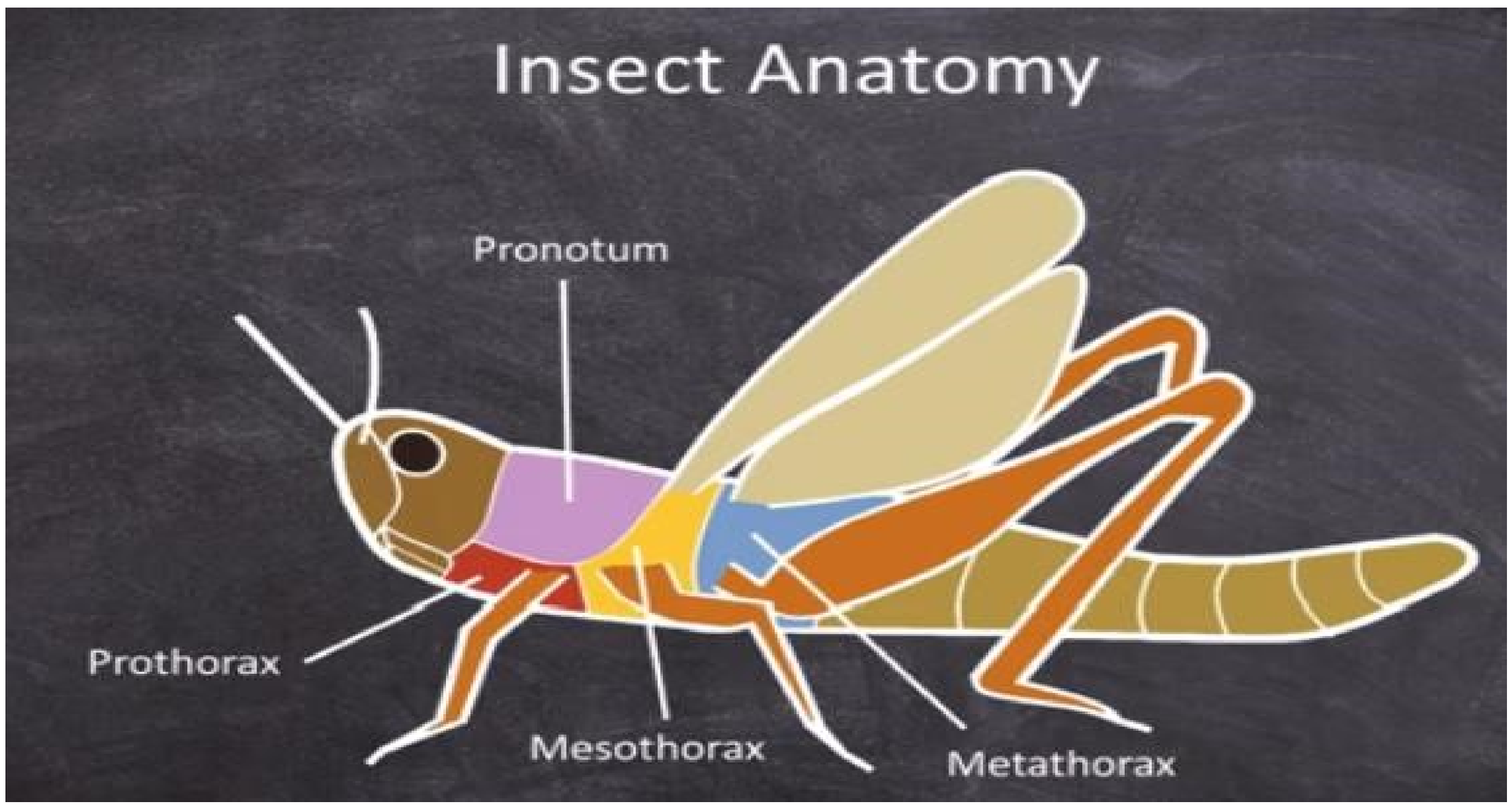
On the second, or mesothorax, and on the third, or metathorax, both wings and legs occur in the
majority of insects
There is a tendency in some groups, carried farthest in the higher Hymenoptera, for the first

Segment of the abdomen to consolidate more closely with the metathorax

Than with the second abdominal segment, which in such cases is often slender and gives

Thereby a semidetached appearance to the rest of the abdomen

As though the line of division between thorax and abdomen were at that place instead of Farther forward



Q: Define ropodeum/ median segment??

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The first abdominal segment, when seemingly more apart of the thorax than of the abdomen, Is called the median segment or ropodeum

21 -22. Structure of legs

The three pairs of legs may be quite similar or differ widely, according to the uses to which

They are put In running and walking insects they are usually most similar But when, for

Example

the forelegs are used for capturing other insects, their form will depart greatly from That of the others why grasshopper has jumping power? ?

The jumping power of the grasshopper is due to the great development of its Hind legs as compared with its others

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Whatever may be the variations in form and details of the legs

All are composed of a definite number of pieces or segments, connected by hinge joints so describe the parts of legs??/describe the structure of legs???

Arranged that, by combining the motions of these a leg can be placed in nearly in any

Position desired The leg is composed of coxa, trochanter (two in a few cases), femur, tibia, Tarsus The last is really not a single segment but a row of from one to five, small, and on the whole Rather resembling each other.

1. The coxa is the segment that articulates with the body, frequently partly lying in a more or less

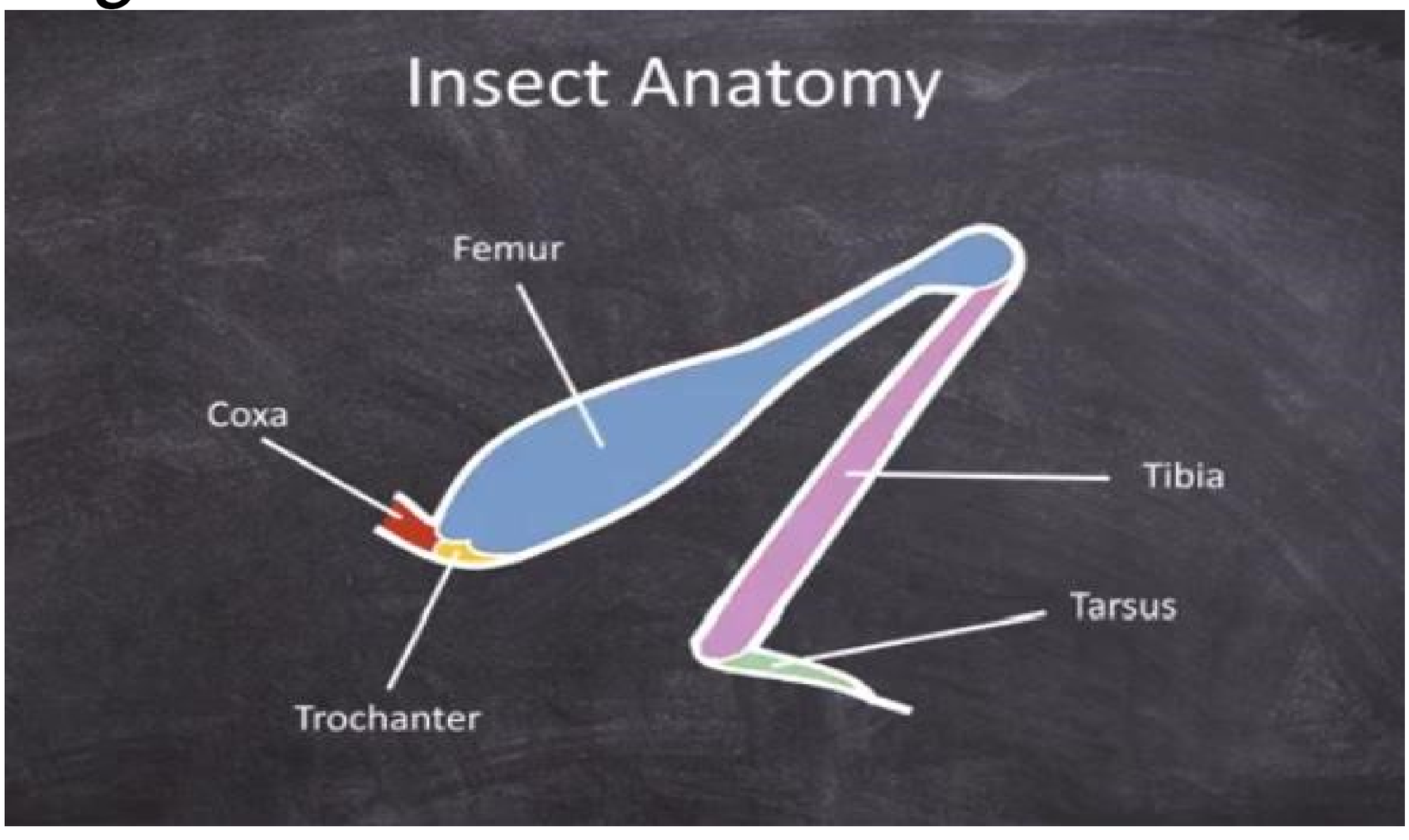
Cupshaped hollow of the latter

It may be short or long and is generally freely movable on the body and powerful

2.The trochanter is usually small and may not be visible on all sides of the leg

3.It is followed by the femur, generally the largest and stoutest, but not often the longest leg

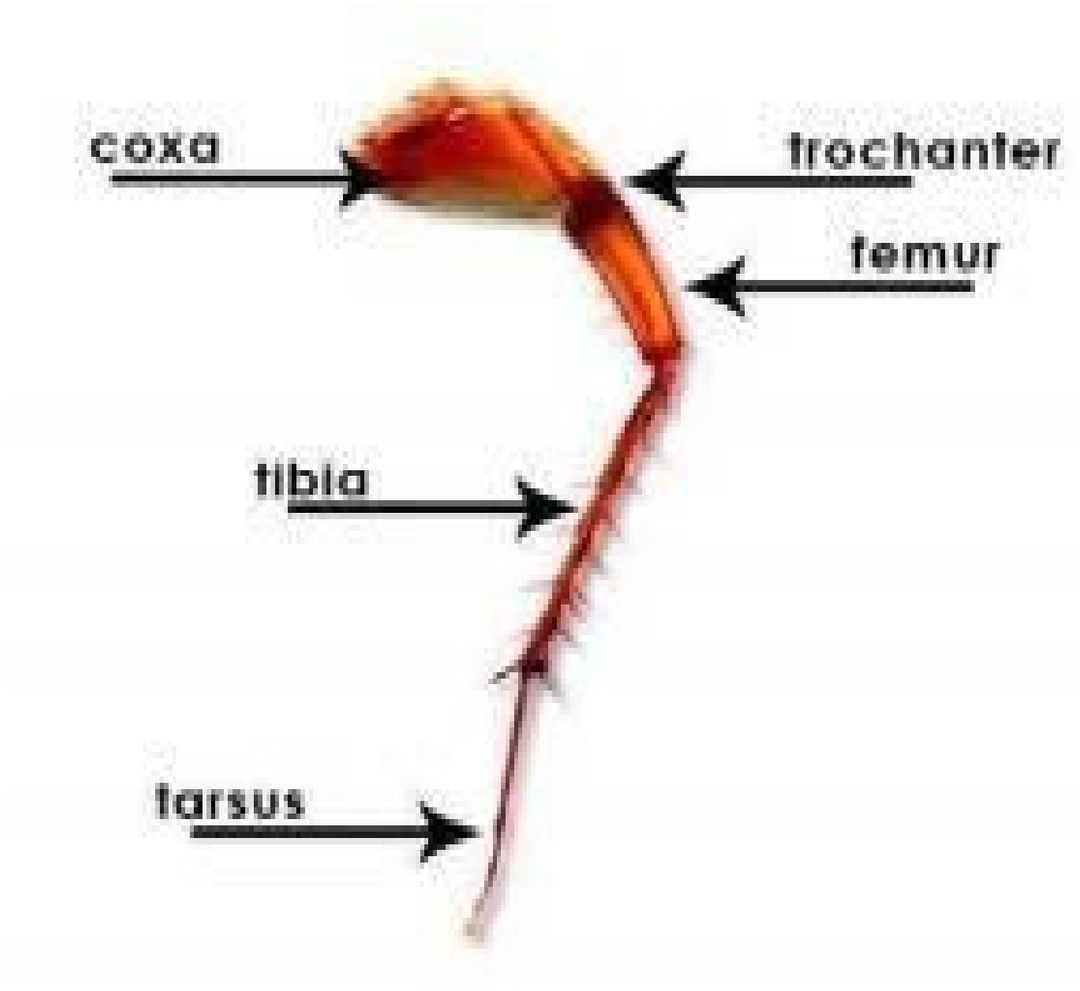
Segment



4.The tibia is in most cases quite long, more slender than the femur and often provided with

Downwardly projecting spines or other structures which are of assistance to the insect in

Climbing plant stems and other objects, to help prevent slipping



5.The tarsal segments are generally rather small, short, tend to be broadest at their outer ends

And vary greatly in details of structure

At the end of the last a pair of claws is generally found, and between them a sort of pad or Cushion, the pulvillus

Sometimes there are three of these, in which case the outer ones are called the pulvilli and the

Middle one the empodium

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Where the tarsi are reduced to a small number of segments, only one claw may be present.

23,24. Structure of wings

describe the structure of wings???

The wings are chitinous outgrowths from the body which vary much in size and form in different insects Each consists of two delicate membranes in contact with each other except along certain lines

Along these lines each membrane thickens and also rises above the general surface So that if the two membranes could be separated and examined from the inner surface, they would appear uniform except for grooves with thickened sides and bottoms, running here and there

When the membranes are brought together again, these grooves combining form hollow rods which, being stronger than the rest of the membrane, serve as its support and hold it stiff These hollow rods are usually called veins or nerves, though they are nothing of the sort.

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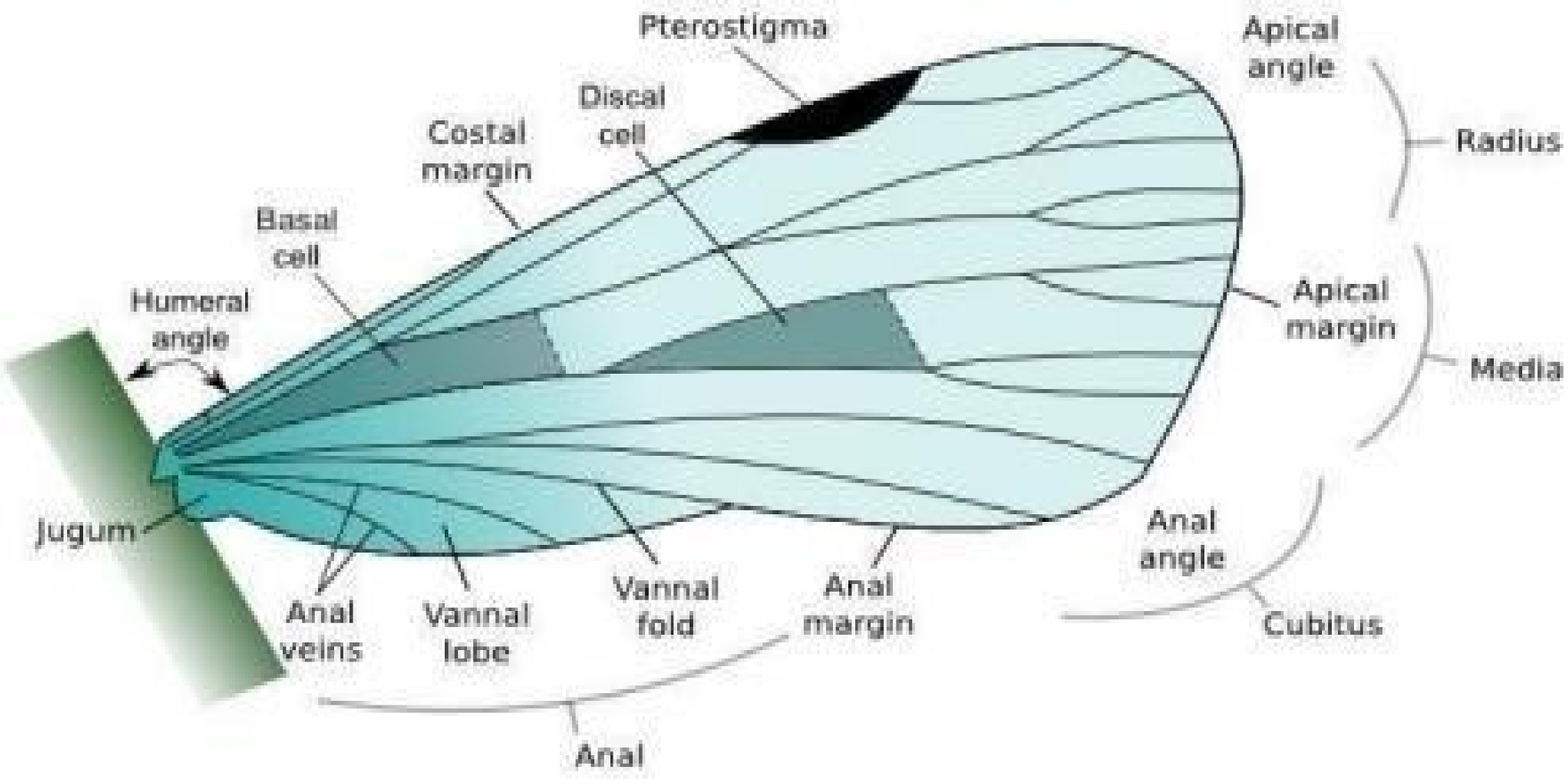
The main and largest veins arise at the base of the wing and extend outward, branching as they go, and some branch several times before they reach the wing margin Cross veins also occur, connecting' the radiating main veins or their branches

Cells?? Areas of membrane between veins are termed as cells and

Closed cells?? where entirely surrounded by veins are called closed cells These may be relatively few or many, according to the number of veins and their branches present The arrangement and number of the chief veins and their branches are of importance in identifying insects

Apex: There is usually a point or tip called the apex,



somewhere along the margin of the wing, though frequently the outline is so rounded that the exact apex is uncertain
Costa?? The front margin of the wing from where it joins the body to where the edge begins to turn backward
(in an extended wing) is called the costa
Wings are entirely absent in some groups of insects It is probable that some of these are direct descendants of the earliest forms before wings were developed In other cases where wings are absent this condition is associated with a parasitic life Where they might be a distinct disadvantage, or with peculiar habits which would render them useless or even inconvenient; in such cases they appear gradually to have become lost

25,26. Structure of abdomen

Describe the structure of abdomen???🥲

The abdomen does not usually show great differences in its segments except in those near the hinder end Halters?? In the flies the hinder pair is modified, forming small structures called halters

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Which may be modified for various purposes Generally a dorsal plate (notum or tergum) and a ventral plate (sternum) are the only two skeletal plates evident in a segment Small openings, usually a pair in each, or at least in some of the segments, are the openings of the breathing organs Generally a dorsal plate (notum or tergum) and a ventral plate (sternum) are the only two skeletal plates evident in a segment Small openings, usually a pair in each, or at least in some of the segments, are the openings of the breathing organs

And these also occur on some of the thoracic segments where they are ordinarily less notice able than on the abdomen

Legs are very rarely present on the abdomen in adult insects but are often found in the earlier stages Q:define ovipositor??

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- *At the end of the abdomen in the females of those insects which lay their eggs within objects is a combination of pieces known as an ovipositor*
- *Ovipositor usually consists of about three pairs of parts, long or short, slender or stout, as the case may be, for the purpose of making a hale or sawing a slit in the object in which the eggs are placed and in guiding the eggs into the hole thus made*
- *In one group which has apparently changed its habits and no longer needs to make holes for egg laying, the ovipositor being unnecessary for this purpose has been transformed into a sting Q:Define the term cerci??*
- *A pair of many-segmented, antenna-like structures, sometimes short, sometimes long, may occur at the end of the abdomen And these are called cerci*

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- *They probably serve as organs of touch and possibly organ of smell in some cases*

27: the internal structure of insects

Q: describe the internal structure of insects??

- *Few of the internal structures of insects are of any great importance from the standpoint of control methods but some knowledge of them and their arrangement is desirable*
- *The alimentary canal extends from the mouth through about the center of the body to the anus at the hinder end*
- *In those insects whose food is most concentrated*
- *It is in its simplest form In those which feed on less concentrated food*

28,29,30: digestive organs

Q: describes the digestive organs of insects??

- *The alimentary canal extends from the mouth through about the center of the body to the anus at the hinder end*

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- *In those insects whose food is most concentrated*
- *It is in its simplest form*
- *In those which feed on less concentrated food*
- *The necessity for a greater digestive and absorptive surface has resulted in an increase of its length And the accommodation of this within the body by the production of loops and coils*

Q: describe alimentary canal of insect's embryo??

In the embryo the alimentary canal forms as three separate sections which connect later

1-One of these is an ingrowth from the surface where the mouth is to be

2-Another and similar ingrowth occurs where the anus forms

3-And a third, forming earlier than the other two, arises as two masses of cells

One near each end of the embryo, which move inward and toward each other, unite, and surround the yolk Later, when this has been absorbed, a space is left with which

the two ingrowths already mentioned connect Fore intestine??? 🤢

The hollow centers of all three joining to form the tube through which the food travels The ingrowth from the mouth is usually called the fore intestine Mid intestine??

The central portion the mid intestine and Hind intestine??

the ingrowth from the anus the hind intestine

The first and last of these begin to grow inward from the surface of the body after that surface has begun the formation of its chitinous exoskeleton And accordingly they also have this power, and line the inside of the parts of the canal which they form, with chitin

In that portion of the canal termed as the mid intestine However, this power does not appear to be present, and the mid intestine is without this lining The mid intestine forms the stomach of the adult insect

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- *The fore intestine forms those parts of the alimentary canal from the mouth of the stomach*
- *Thus the fore intestine, by differences of structure, may sometimes consist of*

    - *Mouth cavity*
    - *Esophagus*
    - *Crop*
    - *Proventriculus*

- *The hind intestine those from the stomach to the anus Each of these sections may sometimes have portions differing in structure, producing a greater or lesser number of subdivisions*

The stomach may develop side pouches or gastric caeca And the hind intestine is often separable by differences of structure into an

- *Ileum , • Colon*
- *Rectum.*

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Lined as these parts are by chitin which often bears rough, tooth-like projections and spine Some persons have suggested that in insects where these structures are present in the fore intestine

The food is masticated more thoroughly and mixed with digestive juicer before it reaches the stomach In the stomach, digestion is probably completed and absorption at least begun

But the length of the hind intestine in many insects suggests the idea that absorption in those cases has not been completed when the food leaves the stomach but continues in the hind intestine

Opening into the mouth is a tube leading to the salivary glands

Which generally lie in the front of the thorax and appear to have a similar function to those in man

In some cases other glands for different purposes "" are also present in the head or front of the thorax and open into the mouth

Stomach poisons??

- *Some of the poisons used in control measures are swallowed by the insert,*

passing to the stomach and there are dissolved by the digestive juices

- *Thus dissolved, they set up inflammation of the stomach walls and finally cause death Poisons acting in this way arc called stomach poisons*

31,32: Breathing organs

Q: describe the breathing organs of insects??

Respiration in insects is accomplished by a method which is nearly unique
The oxygen needed, instead of being ,drawn into lungs and there being taken up by the blood and carried to 'the parts of the body Where it is needed, as in man, is carried directly to those parts by a system of air tubes which open along the sides of the body Here the air enters the tubes and proceeds through them to where it is utilized

Spiracles??

- *The openings by which the air enters are called spiracles, and these occur in pairs on some of the thoracic and most of the abdominal segments, • Varying somewhat*

in number and in position “u the segment in different insects

- The spiracles often have valves which they can be more or less completely closed at will
-

Q: describe the trachea / tracheal system of insects??

- Each spiracle opens into a short tube or trachea which, with the others of that side, soon joins a similar tube running along the side of the body and quite close to its surface
- From these longitudinal tracheae, branches pass off in various directions and in turn branch again and again until every part of the body is reached by its air supply
- The tracheae frequently enlarge here and there, forming so-called **air sacs**
- The tracheae are lined by chitin connected with that of the surface of the body
- In these tubes, however, it is formed with spiral thickenings which act like a spring, keeping the tracheae open when not under pressure

- *There is probably considerable pressure on them in different places by the movements of various parts of the body in walking and other activities*
- *As well as by regular respiratory movements, and the resulting temporary variations in diameter aid in the circulation of air in these tubes*
- *Not only are the tracheae of use in: carrying oxygen to all parts of the body But they also receive much of the carbon dioxide gas produced by the activities of the cells and permit it to escape through the spiracles from the body*
- *Thus performing both of the functions which the blood, so far as gases are concerned, accomplishes in man Blood then, in this gas enters the spiracles and follows along the tracheae to the living tissues, which take it in place of the oxygen usually received in this way, and the insects are killed*

Q:what do you know about contact insecticides??/define contact insecticides??

It was formerly supposed that certain materials called contact insecticides, which kill insects by contact with their bodies
Caused death by entering the spiracles and closing them up, thus producing suffocation.
This has now been proved to be incorrect in most cases
Tracheal gills???
Insects which in their early stage of live in water cannot, of course, breathe air into their bodies through spiracles during that period of their live
There are enclosed in such cases and the animal obtains and usually through special structures called tracheal gills
These will be described in connection with the insects that possess them
In a few small water inhabiting forms, the chitin covering the surface of the body is so thin that oxygen present in the water can pass directly through it into the body and to the parts there that need it, and carbon dioxide passes in the reverse direction,

33,34,35- Circulatory organs:

Q: describe the circulatory organs of insects??

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Insects have only an incomplete system of blood vessels A tube lies in the middle of the body close beneath the back, beginning near the hinder end of the animal and extending forward into the head

Q:write a note on the heart of insects??

Heart:

In the abdomen this tube is constricted, forming chambers, and the chambered portion is called the heart There is a pair of openings on the' sides of each chamber through which blood can enter, and valves there which prevent its going out again In the abdomen this tube is constricted, forming chambers, and the chambered portion is called the heart
There is a pair of openings on the' sides of each chamber through which blood can enter, and valves there which prevent its going out again Blood in the heart, being unable because of the valves to pass out at the sides, is pressed forward by this contraction wave And at the front end of the heart finds itself in a tube without chambers or valves, called the aorta Through which it is led to the head where the aorta may divide into a few short branches

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or may be unbranched In either case, at this point the blood pours out of it into the body, the system of blood vessels coming to an end There is now no definite and particular path for the blood to follow But it would, in theory at least, remain near where it escaped from the aorta or gradually pass into any spaces it might find unoccupied between the different structures in the head

Q: What happened with the each heart beat in insects?/pathway of blood circulation in insects??

With each heartbeat, however, more blood is poured out of the aorta, increasing the pressure upon that already in the head It therefore is gradually forced backward and to other parts of the body

Each particle probably taking the path where there is least resistance to its passage In this way a general backward direction is given to the flow As it approaches the heart, another influence appears During each contraction, the heart occupies less space, which leads to less than normal pressure near it And blood close by naturally flows closer to it Upon its expansion

again and the opening of its valves, the direction of-least resistance is now through the valves and into the heart As the blood passes back through the body, a given particle may at one circuit go over certain organs, and at the next over entirely different ones

All the internal organs, however, have their surfaces bathed by blood And this as it passes over the stomach or other parts of the alimentary canal will pick up any food Which having been digested has passed through the canal walls

Likewise in passing over any organ needing this food, it is given up to those organs The blood therefore serves as a distributor of food from the place where it is digested to all the parts that need it

We have already seen that the living parts of the bodythe cells-need oxygen and as the result of their activities give off carbon dioxide gas

But that this exchange is accomplished by the aid of the tracheae In a somewhat parallel way, the cells that need food obtain it from the blood

The cells by their activities produce not only carbon dioxide gas but also waste material nitrogenous in nature which must be removed like all wastes from the body

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36 37 -Excretory organ

Q: write a note on excretory organ of insects??/

How nitrogenous waste is picked up at the cells

??😪

This nitrogenous waste is picked up at the cells by the

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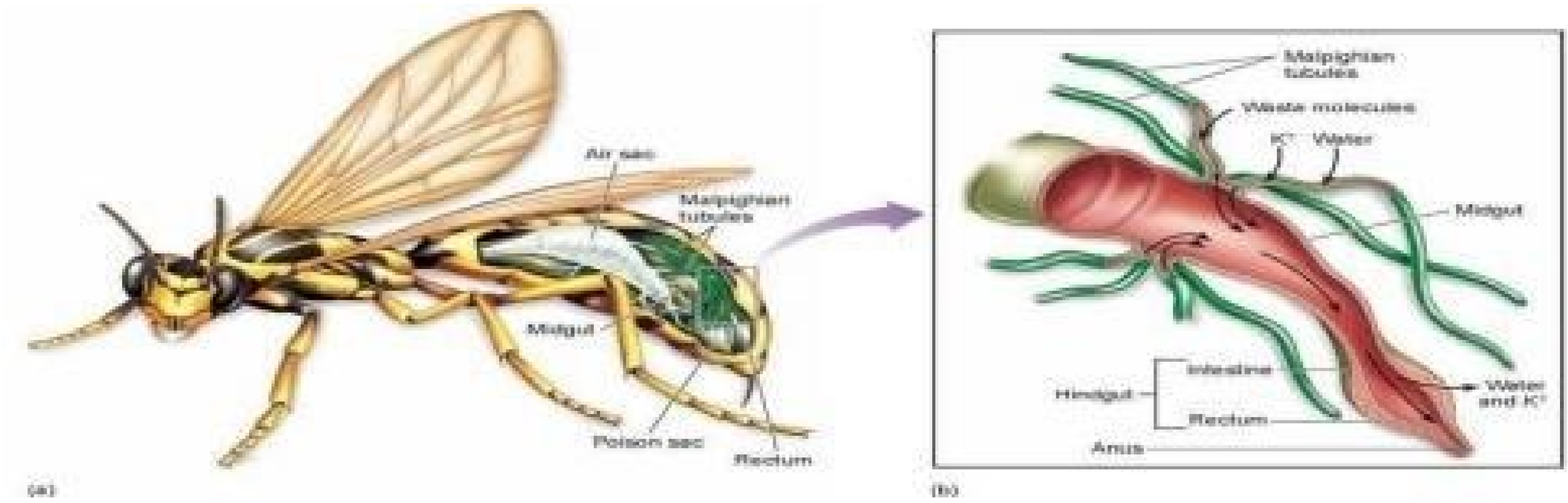
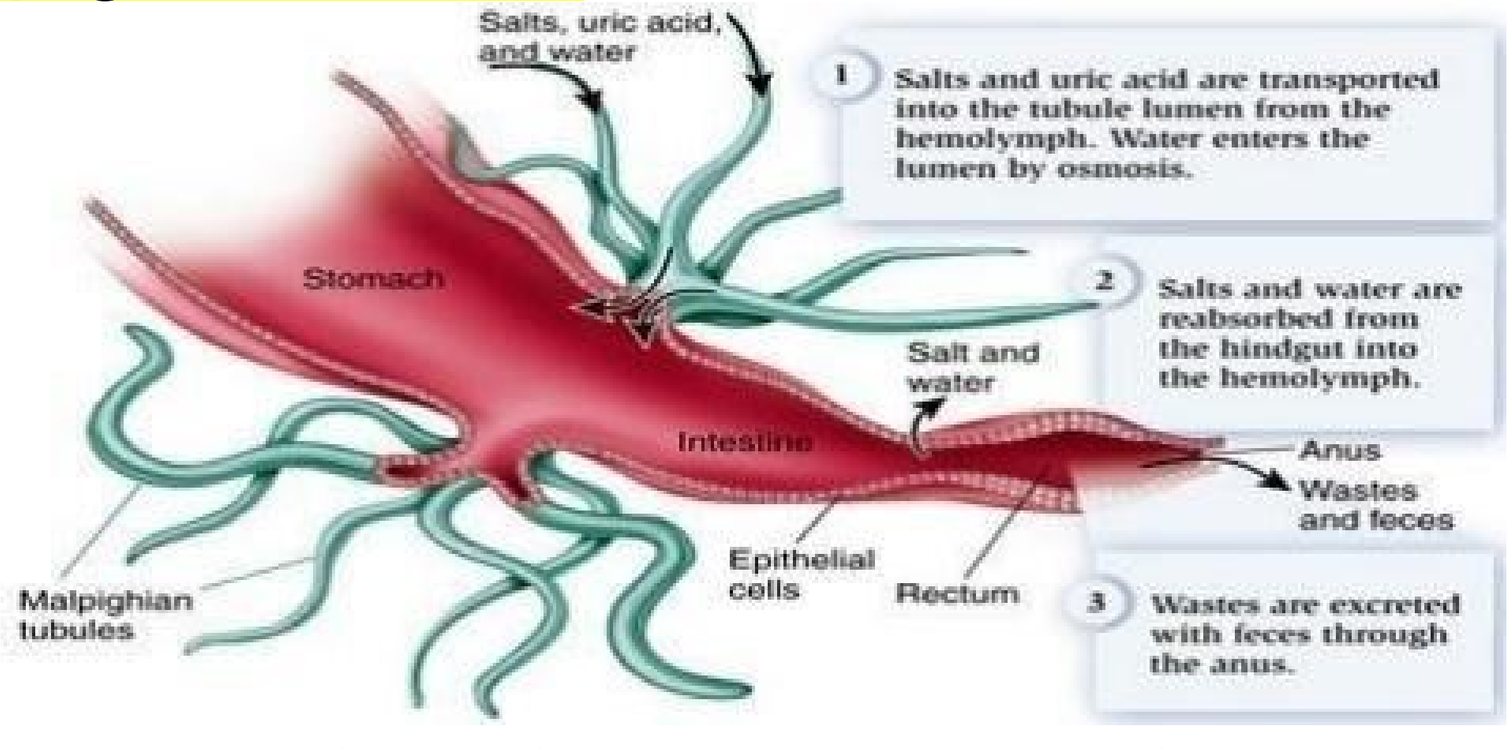


blood and carried along, perhaps for some time, before a place to dispose of it can be found Sooner or later, however,

a particle of blood containing this waste material will wash over certain structures called ***Malpighian tubes,***



to be described in the next section And the cells which form these tubes have the power to collect this waste material from the blood as I it flows over them, thus purifying it

Q: describe the color of blood in insects?

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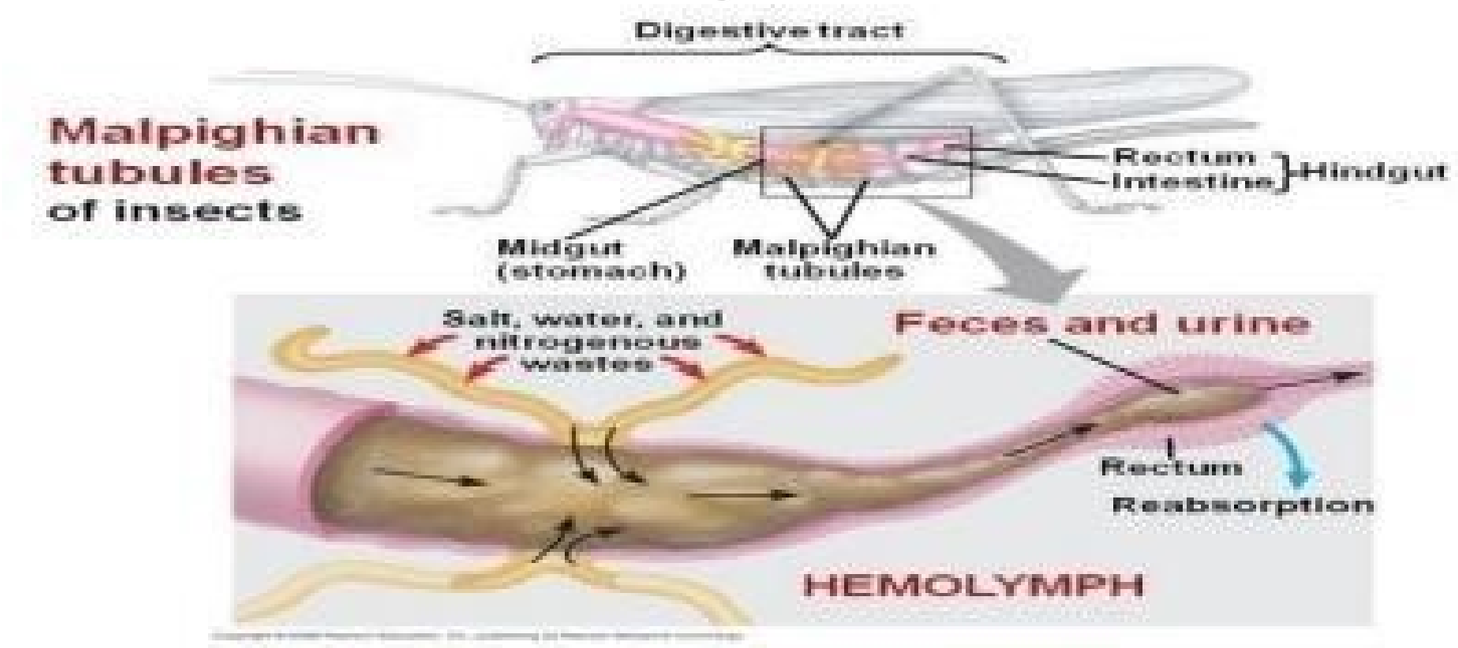
The blood itself is usually a colorless, yellowish, reddish or greenish fluid, in which are corpuscles resembling the white corpuscles of human blood

- *It appears to serve to carry food to the tissues and waste matter from them, and therefore has no need of structures in it like the red blood corpuscles of man*
- *The work of which in insects is done mainly by the tracheae*

Malpighian tubes:

- *The organs that eliminate the nitrogenous wastes from the body and correspond in function to the human kidneys are known as Malpighian tubes*
- *These are blind-ended tubes, the walls of which consist of a single layer of cells surrounding a central channel which at one end open into the hind intestine, Usually near its front, just behind the stomach*



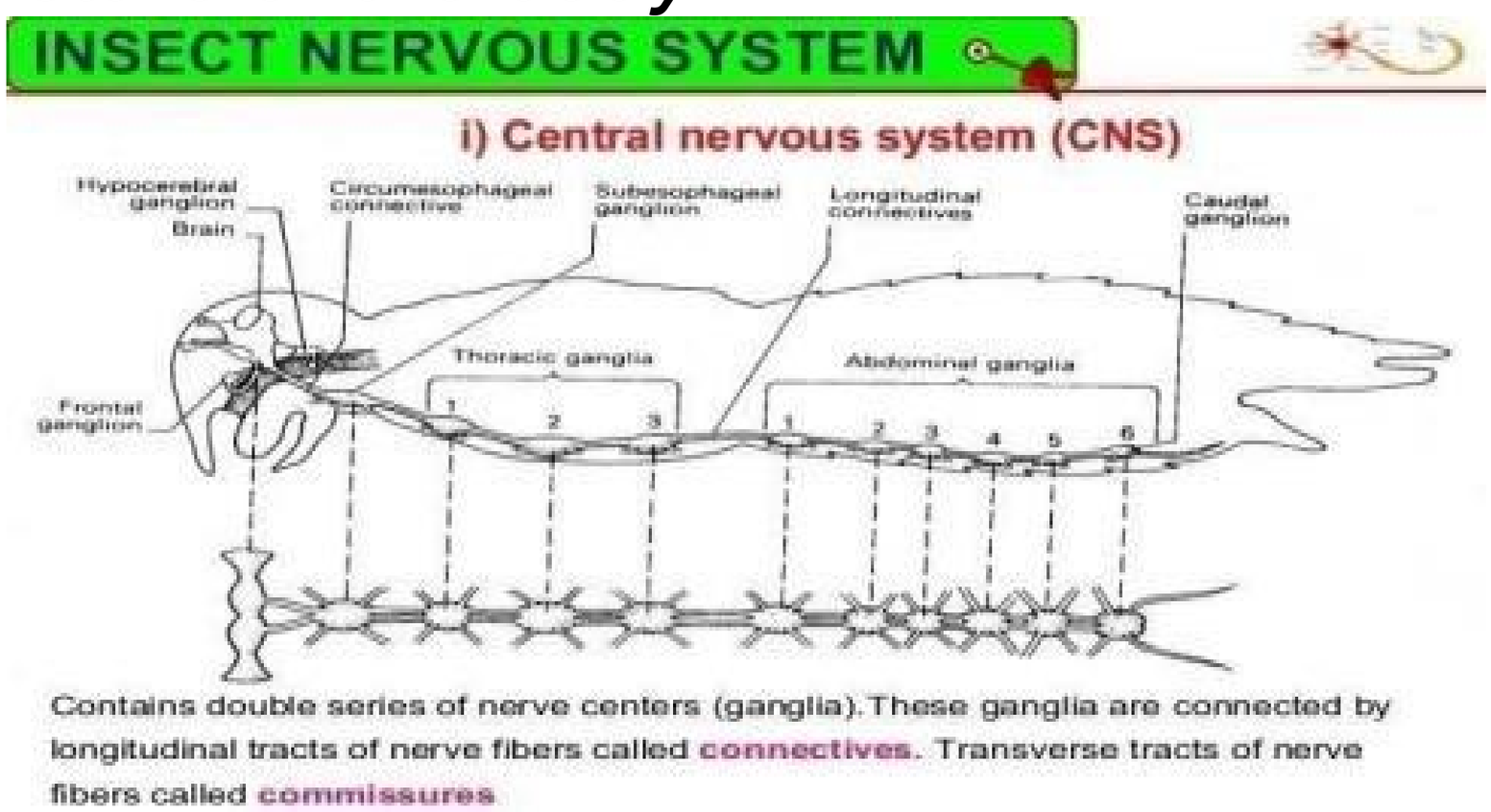
When blood containing nitrogenous waste matter washes over the outer surface of a Malpighian tube The cells of which it is composed have the power of taking this matter out of the blood into their own substance and passing it through them Solves into the channel between them, dowln which it moves until it enter the hind intestine, from which it is finally expelled at the anus
The Malpighian tubes may be few or many, long or short
They show a tendency to collect in groups and to unite near the hind intestine So that their outlets into this are much fewer than the number of tubes.
Q: how the poison get eliminated from the body of insects?
It seems possible that a certain amount of poison entering the body by way of the stomach can be eliminated by the Malpighian tubes
Which may explain the varying degree of resistance to such poisons by different insects

38,39- nervous system

Q: what do you know about nervous system of insects??

The nervous system of insects is located along the middle line of the body quite near its under surface As in animals generally, it is composed of cells and fibers

Ganglia: *former are for the most part gathered together in clusters which are called ganglia And from each of the cells in a ganglion one or more nerve fibers pass out, to connect either with some other nerve cell or with some structure of the body*



The larger nerves are really bundles of the fibers running side by side like the wires of a

telephone cable Apparently each segment of the insect body once had but with the fusion of the segments tube of a fly Greatly many of these have also fused, reducing the separate ganglia in adult insects to a smaller number, which varies central in different kinds
This fusion has been produced by the hinder ganglia
moving forward until in some cases none is found in the abdomen
Q: Define commissure??

Each ganglion is connected to the one in front and the one behind by one or two bundles of nerve fibers which are called commissures Each consists of numerous fibers which taken together form the means of communication between the different parts of the system ***Q: give so idea about the brain of insects??***

In the head, in front of or above the esophagus, is the largest ganglion of the body, called the brain, produced by the fusion of several ganglia In addition to its two commissures, which connect it with the ganglion next behind, it has nerves which lead to the eyes, to the antennae and to other parts of the front of the head Below or behind the esophagus is a second ganglion,

also in the head, called from its position the sub esophageal ganglion As the esophagus lies directly between this and the brain, the commissures connecting the two do not lie close together but separate far enough to permit the esophagus to pass between them The sub-esophageal ganglion, besides being connected with the brain in front And the first thoracic ganglion behind it, by commissures, sends nerves to the mouth parts and other near-by regions of the head. The thoracic ganglia may be more or less separate or fused and may have fewer or more of the abdominal ganglia added commissures However, connect all separate ganglia, and these also send out nerves to all the parts of the segments to which they belong, no matter what their final location may be

In this way, the wings, legs, muscles and other parts receive their nerve supply A small" sympathetic nerve system" also present, appears to concerned chiefly with the nerve supply of the alimentary canal tracheae

40- Sense organ

Q: write a note on sense organs of insects?

? All the more evident senses possessed by men appear to be present in insects
Except possibly taste, but not in all cases in the same individual
Thus some cave-inhabiting insects have no eyes
It is almost certain also that insects have other senses not possessed by man

41 Reproductive organs

Q: write a note on reproductive organs of insects??

Insects are of distinct sexes, male and female *Q : Define neuters??*
In many cases, however, individuals occur incapable of reproduction, their sexual organs not having become fully developed Such inserts may be termed neuters Most of these appear to be really undeveloped females, though undeveloped males are also known

They are found in colonial insects where division of labor occurs

As in the honeybee, ants and termites, and are known according to their duties, as workers or soldiers or by other names

Q: write the conventional signs for male and female insects??

Conventional signs for the various forms of insects, as a convenience, are if male ♂ female ♀ or worker

42,43- Female reproductive tract

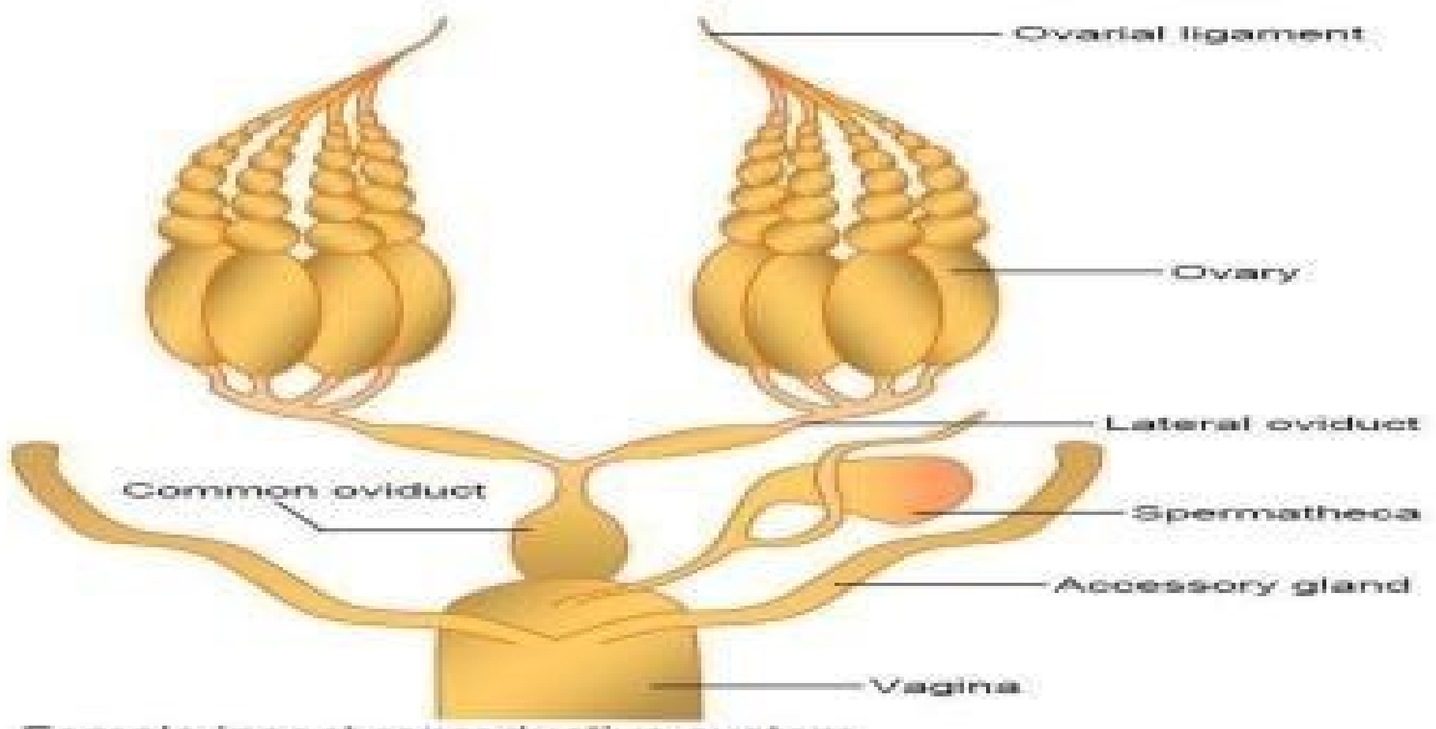
Q: write a note on female reproductive system of insects??

In the female the eggs are produced in a pair of ovaries located in the upper front part of the abdomen

Q: define oviduct??

Each is a cluster of ovarian tubes whose walls are cells Some of these cells grow and separate from the others to lie in the central cavity of the tube and then pass downward Growing till they reach its hinder end Which connects with the similar ends of all the ovarian tubes of that side to form a single tube called the oviduct

This extends downward and back ward around the side of the alimentary canal Below which it joins with a



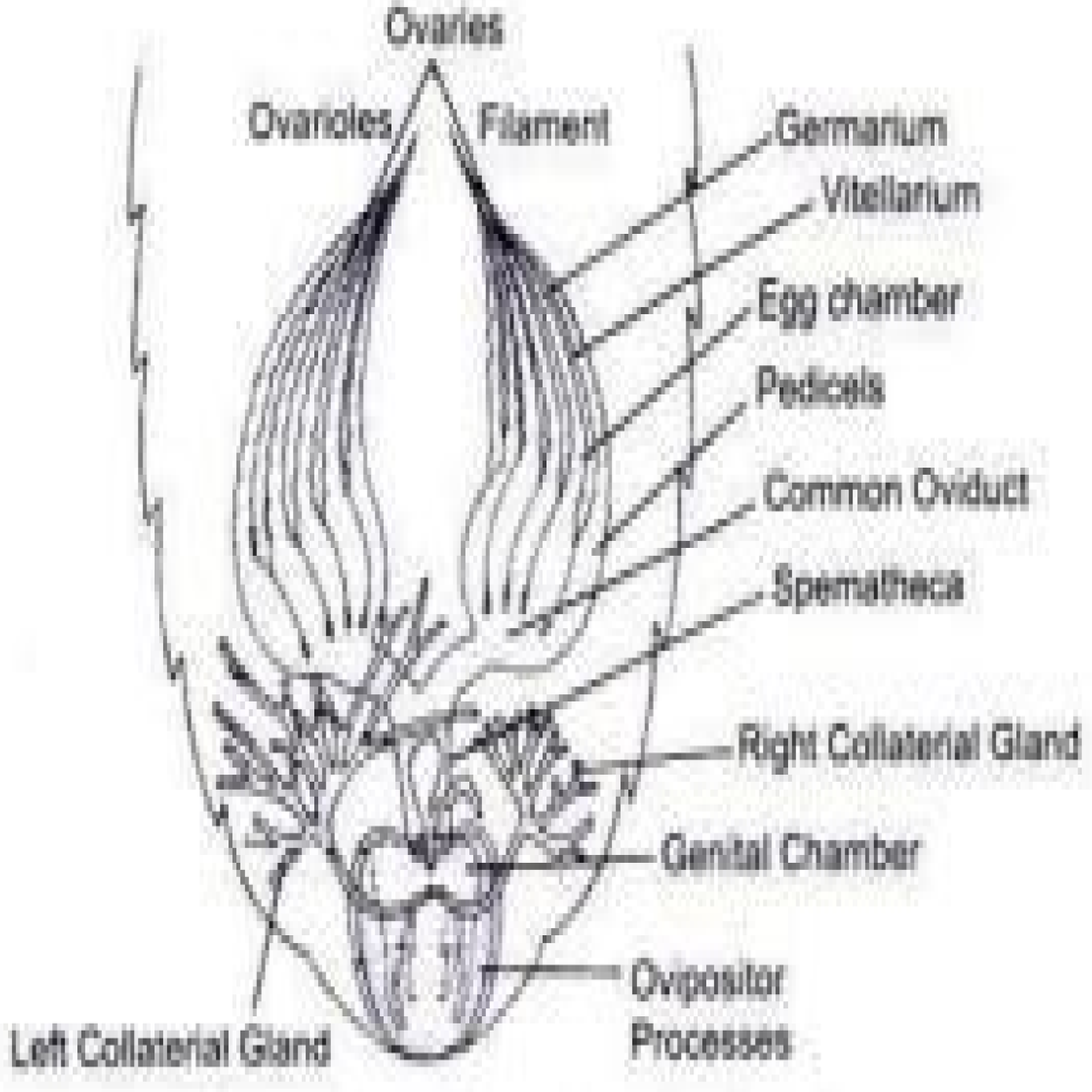
Female insect reproductive system

similar oviduct from the other side of the body to form a single duct The vagina, which lies between the alimentary canal and extends backward to its outer opening which is located In most cases, in front of the next to the last abdominal segment. **Q: Define ovipositor??**

Surrounding this opening may be external structures known as ovipositor For the purpose of together making holes in some object (the ground, wood, etc.) in which to deposit the eggs

Q: Define seminal receptacle??

A side pouch called seminal receptacle is connected with the vagina is for the storage of the sperms which fertilize the eggs



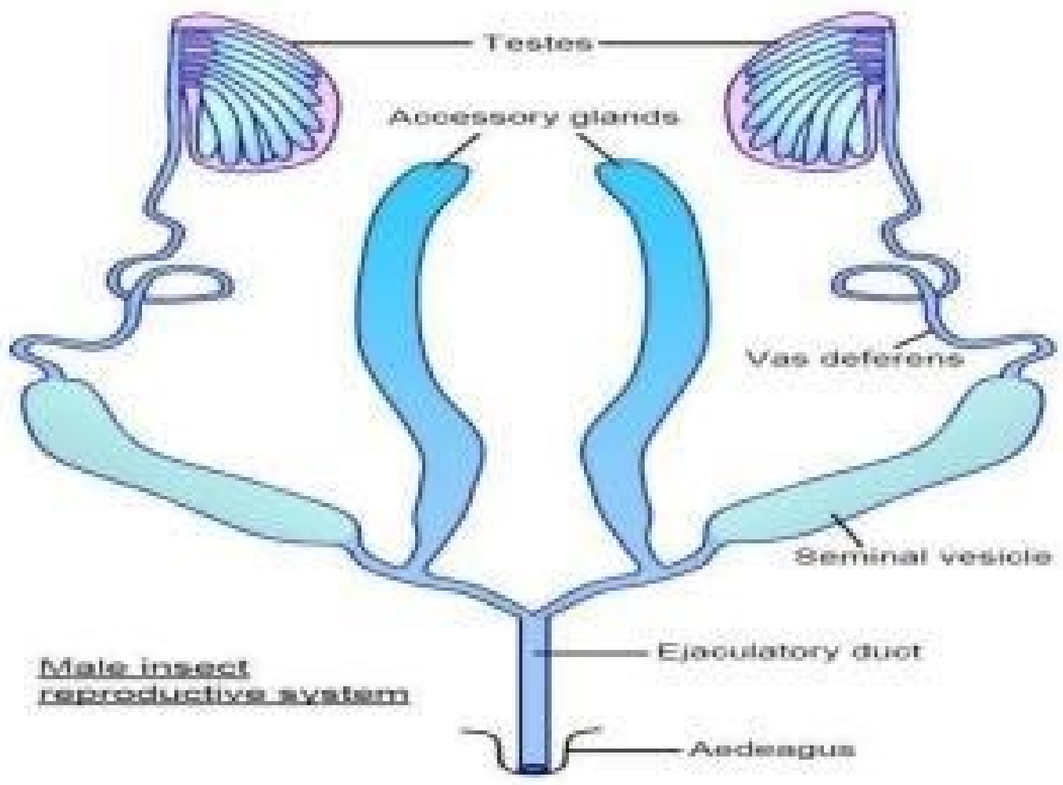
Q: Define shell gland??

A gland-producing material, which forms the egg shell and is known as the shell gland, also opens into this portion And other glands similarly connected with the vagina may also be present.

44- male reproductive tract

Q: write a note on male reproductive system of insects??

In the male the arrangement of the organs closely corresponds to that in the female
A pair of spermaries or testes is present in the upper front part of the abdomen
Each consisting of a rather closely coiled mass of tubes in which the sperms are produced.
The tubes on each side unite to form a single tube, the vas deferens (plural, vasa deferentia)
These differ from the oviducts usually, in being much longer and coiled or twisted ***Q: ejaculatory duct??***
They pass downward and backward, however, and unite on the middle line of the body below the alimentary canal Forming a single tube, the ejaculatory duct,



Male insect reproductive system

corresponding to the vagina in position Which leads backward to an opening in front of the last segment ***Q: seminal vesicle??***

An enlarged portion of the vas deferens is often present, for the temporary storage of the sperms, and is termed the ***seminal vesicle*** *Accessory pouches opening into the ejaculatory duct appear to be, in part at least For the production of mucus and secretions to mix with the seminal fluid*

45- the development of insects (egg types)

Q: write a note on development of insects/ egg types

of insect?!?

Most insects lay eggs from which the young hatch after a longer or shorter time Though in some cases the egg appears to be retained within the body of the parent until after it has hatched The young then being produced in an active stage

Q: Oviparous??

Those which lay eggs hatching later are termed oviparous *Q: ovoviviparous??*

Those having eggs that remain in the body of the parent for a time after hatching are termed ovoviviparous

Q: viviparous??

A few insects that supply nourishment to the young before its birth are termed viviparous

Insect eggs are usually very small, vary greatly in form and may be laid singly or in clusters.

4 structure of Eggs

Q: explain the structure of eggs of insects??

Chorion??

They are covered by a chitinous shell, the chorion, which often bears markings in the form of ridges, reticulations And frequently are also colored

Q: Define micropyle??

At one place on the surface is a minute opening or group of openings through the shell, called the micropyle Believed to be for the entrance of the fertilizing sperm

The length of time spent in the egg differs in different insects from a few hours to many months, and in some cases the eggs Do not hatch until the second season after they are laid

In hatching, the shell breaks and out of it crawls the young insect

Q: what do you know about insect's metamorphosis ?

In the majority of cases quite unlike the adult it is to become In order to reach maturity it must now grow and undergo changes in structure and appearance These together are expressed by saying that most insects in order to become adult undergo a metamorphosis

47- Metamorphosis

Q: types/ forms of metamorphosis??

In some of the simpler inflects, a few changes and growth only are needed to make them mature *Q: Ametabola??*
And these are therefore usually grouped together as the
Ametabala, or insects having practically no metamorphosis
The remaining insects, from this standpoint, form two groups

Q: Define Hemimetabola/ paurometabola?

Hemimetabola, or
Paurometabola:

Those which on hatching generally show some resemblance to the adults and reach maturity by a series of gradual changes

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Q: Define holometabola?

Holometabola: And those which on hatching are totally unlike the adults and become adults by a somewhat different process 🦋

These names suggesting the amount of metamorphosis required for members of each group to become adult A member of the group Ametabola upon hatching, will begin to feed and grow Growth, however, is restricted because the insect is enclosed by a cuticula which, while elastic, to some extent, at least at its thiner places, has its limitations in this regard Thus the insect is unable to reach its adult size within its cuticula, and a process called molting takes place This is begun by a pouring out of fluid by the outside layer of living cells, the hypodermis, between it and the cuticula, separating the two Next a spilt appears somewhere in the cuticula usually along the back and the insect crawls out of its skin, i.e., molts.

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48-50- Types of Metamorphosis

Until the fluid poured out has had time to harden, the insect is able to grow But in a short time it hardens to become a new cuticula and there after only such growth occurs as the elasticity of the new shell will permit In theory there should be some Ametabola that do not molt, the cuticula being so elastic it will stretch enough to permit growth to the adult In fact, however, they· all appear to molt at least two or three times in a few cases it seems to occur at intervals throughout life In any case the reproductive organs appear not to be mature at the time of hatching and only gradually become so during the period following In the Hemimetabola (or Paurometabola) the young insect on escaping from the egg, though resembling its parent to some extent Undergo many change in structure and a considerable increase in size as well before reaching maturity Thus a young short-horned grasshopper, on hatching, will need to grow to be about ten times as long before becoming adult

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It is without wings, which will need to be developed; its reproductive organs are not mature and must become so; and other differences occur
All of these must be transformed into their condition in the adult; and to accomplish this, energy is necessary In the egg the energy for development has been provided by the yolk; after hatching the young insect must provide it by gathering food
The young insect, therefore, soon after hatching seeks for food and haring found it begins feeding
The nourishment thus obtained results , in growth so far as this is possible within a shell which is tightly fitting and only to some degree elastic
When no further growth in this way can occur and the body has stored within it all the materials needed for a greater increase in size It proceeds to molt in the manner already described for the Ametabola On escaping from its old skin or shell, how ever, besides a rapid increase in size, changes of structure also occur So that a difference in appearance now becomes evident These changes must be produced quickly, as the hypodermal cells of

these parts, as well as of all the surface, are producing a new chitinous skin When this has once hardened, no further changes and little further growth are possible Molting, then, marks the beginning of a brief period-a day, more or less-of increase in size
and of changes in appearance

These last all being in the direction of making the young insect more nearly like the adult it is to become When the new shell has become hardened, the insect resumes its feeding After another feeding period the young insect is again confronted with the same difficulties as before, and it meets them in the same way, by molting, and immediately thereafter Before its new shell has hardened, it seizes the opportunity to grow and change its appearance further Finally, after some molt, full adult size for the insert is attained and all its organs have also fully developed and matured, producing the adult insect itself

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51-53- - Hemimetabola

The number of molts, and consequent opportunities for change that occur, vary in different Hemimetabola There may be only two or three in some kinds; five is perhaps the average number though more are not uncommon; forty-five are known to occur in one species Certain names for these different conditions are convenient for use Q: Define instars??

The feeding periods between the molts (or ecdyses) are called instal's So that the progress of an insect from hatching to adult is by an alternation of instal'S and molts

Q: Define numph??

The insect itself, from hatching until maturity, is generally called a nymph

The changes in appearance of a grasshopper after each molt

Q: write the orders of Hemimetabola??

The term Hemimetabola was used for many years to include all those orders in which the insects underwent incomplete metamorphosis in the course of their

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development Found in In three of the orders 🐒

1-Ephemerida

2-Odonata

3-Plecoptera

In three of the orders, the young live and develop in water and differ greatly from their adults in appearance In recent years this has led some entomologists to divide the Hemimetabola into two groups The three just named were left under the Hemimetabola and the others were given the group name Paurometabola which means a gradual metamorphosis Q: Define naiads??

The young in the Hemimetabola thus limited are called naiads; those in the Paurometabola, nymphs This is by no means a classification of the insects themselves, but only of the different types of develop meant that occur This classification has not been adopted in this book. With the remaining group of insects, the Holometabola, although there is a little similarity in the metamorphosis to that in the

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Hemimetabola But there are also many differences. When a young holometabolous insect hatches, it in no way resembles its adult A caterpillar is totally different in appearance from the butterfly it finally becomes the white grub in the earth It is called June bug (May beetle)?? into which it transforms Nevertheless, this young insect-called a larva has to meet the same problems of growth and transformation to the adult condition as do the Hemimetabola And uses the same means for accomplishing the needed results, · viz., the utilization of the energy derived from its food Upon hatching, in the Holometabola, a feeding period or instar comes first, followed by a molt and growth At this point the story of the metamorphosis differs from that of the hemimetabola, for after the molt no change in appearance to make the young insect more nearly like the adult takes place It may be different in some regards, besides size, from what it was before the molt, but these differences do not increase its resemblance to what it finally becomes

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This holds through out the feeding period of its existence, so that after three, four or more molts a caterpillar is still caterpillar, a grub is still a grub, and this is Equally true for all holometabolous insects With in the insect during this period, however, changes not perceptible on the surface are taking place, by the construction of portions of the adult which are forming as buds or ingrowths from various parts of the body and are termed imaginal buds (from "imago," the adult)?? They are closely compacted and many at least are enfolded somewhat like buds, becoming finally ready to open when the proper time comes And during its feeding instars, the larva is storing energy from its food not only for its growth at each molt but also to carry it on through a period yet to be described During which it must transform into the adult condition while unable to feed and obtain the energy needed for this purpose 54-58.. Development in insects

After a varying number of feeding instars and molts, the young insect or larva has grown sufficiently and has stored within it energy Enough to carry it through the remainder of its changes, and internally the essential parts for

the adult condition have been formed as far as it possible under existing conditions
As the next change will produce an animal practically helpless in most cases and unable to protect itself from

its enemies, its next step is to find as much protection as possible Accordingly, the full-grown larva usually, though not always, leaves the place where it has been feeding and elsewhere prepares for its next change Many larvae begin this by spinning around themselves a thread of silk, produced by glands within the body and opening to the surface on the lower lip This thread is spun backward and forward and around the body until it sometimes forms a complete outer covering, entirely concealing the larva within from view. This case or cocoon appears to be protective in its function

Some larvae go underground for this change
Here a cocoon, as such, seems unnecessary
But after digging into the earth a few inches, the insect forms a little earthen chamber or cell in which to lie and generally lines this more or less densely with silk, probably to keep the earthen walls from falling in and crushing it
A larva transforming in tunnels in wood where it has fed may make a partial cocoon with more or less of the chewed wood fragments mixed in
One staying above ground but not in tunnels or otherwise protected will spin more or less of a cocoon as already described The completeness of the cocoon, however, varies greatly with different insects
Instead of being a thick, dense wrapping which entirely conceals the insect, it may be so scanty that the animal within can be seen to some extent
In other cases it is merely a sort of network, in no degree giving concealment and in still others, a few scattered threads to hold the insect in place are all that represent it Sometimes hairs from the body of the larva, held together by silk, form most of the cocoon

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In the case of butterflies only threads enough to attach the hinder end of the body at the place where it is to transform and usually to form a supporting loop around its middle
The ends of the loop also being fastened to what it rests on, are produced

Q:Puparium??

In some flies the larva shrinks within its larval skin and transforms, this skin, now called a puparium, functioning like a cocoon The reason for such variations in a structure, presumably formed for the purpose of protection, can only be guessed at Possibly in the course of generations some insects found less need of this than others and gradually reduced it Thereby saving the vital energy so much needed for transformation, which would otherwise be expended in cocoon making Whether the larva forms a dense or scanty cocoon, or none whatever, the next step in the process is a molt

Q: Define pupa??🙈

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When the insect escapes from this skin, however, a great change in it appearance is evident, and it is now called a pupa
In a general way it may be said that it has at this one molt changed more than halfway to its adult condition
This is due in part at least to the unfolding of the imaginal buds already referred to Which contribute largely to form the new surface of the body in which Head Thorax Abdomen are evident Antennae, Legs, stubs of wings Many of the internal organs of the larva, though, were necessary for use till the last moment before it became a pupa Then, too, the arrangement of the muscles in the larva would not be that needed by the adult Accordingly, most of the internal organs now gradually break down, losing all their earlier form and structure and new ones to meet the needs of the adult are constructed to take their place
During this breaking down and the reconstruction period, the pupa is practically helpless in most cases; hence generally the need for the protecting cocoon or earthen cell it constructs
When the structure of the adult insect has been completed, another molt takes place, the pupa

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skin splitting and setting free the insect If it was enclosed in a cocoon, it now produces a fluid which sufficiently softens the silken threads so that it can push its way out and it escapes or "emerges." It is now soft its wings are only partly expended as in most cases there would be no room for full-sized wings in a pupa, and because of its reconstruction there is considerable waste matter in its body

The Insect crawls upon whatever it may find to hold on to, expels the waste matter, and its wings begin to grow rapidly

Drying out also takes place and in a short time (a few hours) the adult thus produced is in every way fully matured In some cases maturity of the reproductive organs is not complete until a little later

To summarize the differences in metamorphosis of the three groups it may be said that in the Ametabola The insect hatches from the egg practically in an adult condition; i.e. , there is little or no metamorphosis In the Hemimetabola the insect hatches from the egg in a form somewhat resembling the adult (except in the Ephemerida, Odonata and Plecoptera) but much smaller It becomes adult

by alternating periods of feeding with molts, at which times growth and changes bringing it nearer
To the adult occur, the last molt completing the growth and adult structure
In this life history we have a change; but as there was a resemblance to the adult from the start, the change to it (metamorphosis) is only an incomplete or partial one In the
Holometabola the insect hatches from the egg in a form totally unlike the adult
Where as feeding periods followed by molts and growth give increase in size, no external evidence of any changes making the insect more like the adult can be found.

59.. Development of insects

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Q: write down the development in insects??

These changes are largely made after the end of the feeding and growing periods during a pupa (generally quiet) stage

In which the breaking down of the larval, and construction of the adult, structures is completed. The difference between the larva on hatching and the adult is so great that an entire change (complete metamorphosis) takes place
It should be evident from the foregoing that, when the adult condition is once reached If any growth is possible and that the belief so common that "big flies grow from little flies" is without any basis of fact
The nymphs of the Hemimetabola appear not to have
attracted sufficient attention to receive any special common names
In the Holometabola the larvae of various groups differ greatly in appearance many are large and noticeable and some of them have, as a result, received special names.
Types of larvae:

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Larvae of butterflies and moths are commonly called caterpillars
Those of beetles are usually called grubs
Those of flies are called maggots

. Q: what do you know about borers??
Larvae found boring in wood, however, whether they will become moths, beetles or other insects, are uniformly called borers.

Q: write the stages of life in hemimetabola??
In the Hemimetabola, then the

stages of life are

- *Egg*
- *Nymph*
- *Adult*

In the Holometabola they are

- *egg,*
- *larva, • pupa, • adult*
- *Whether or not the pupa is enclosed by a cocoon depends upon circumstances*

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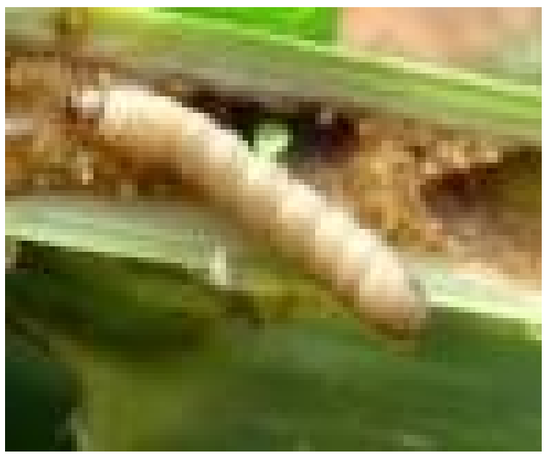

60-61-62-63-64 ECONOMIC IMPORTANCE OF INSECTS (INSECT DAMAGES)

Introduction

A. Of the many thousands of insect species known to man some affect him and his Property only slightly; some are beneficial; others are injurious in various degrees

B. Only rough estimates can be made of the monetary losses and gains by insect activities And even these are made with difficulty

C. Although it is the general practice to express losses in terms of dollar valuation D. It probably would be safer to give them in bushels, tons, etc., because the relative Values of different crops and products change from year to year

E. No crop of either vegetable or animal origin appears to be entirely free from insect injury

E. Field, truck and fruit of crops, both growing and in storage, household goods and food

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Products, forests and the wood products derived from them, domestic animals and Animal products, all are more or less liable to insect attack

F. Because there never has been a season free from the ravages of insects which mightServe as a standard for comparison, much of this loss is not appreciated

G. If there could once be such a year entirely insect free, the difference would be apparent At once

Q write a note on INSECT DAMAGES ??

A. It is the general conclusion that the loss to all sorts of crops by insect damage in an Average year is about 10 per cent

B. This estimate covers field crops, forests and forest products, farm wood lots, Domestic animals and their products, stored articles, shade trees and ornamental Plants, household goods and foods.

C. Generally the injury to fruit and truck crops is believed to be more than one tenth

D. Hyslop estimated in 1938 that the annual loss to agricultural crops by insects, Including the costs of control, amounts to somewhat over 1,600 million dollars

E. No estimate of injury to human health is included

F. To value this destruction we have as a criterion only the price for which crops sell

G. If the tenth of all crops destroyed had been saved, the price of the whole might have been no greater than it was for the nine-tenths actually produced

H. The loss by insect injury usually far exceeds the actual quantity consumed.

I. Insect-contaminated food products originally intended for human consumption often must be destroyed or used in stock feeds

J. Blemishes in fruit caused by insect bites impair the sale value much more than is accounted for by the actual loss in substance.

Plant Diseases

• The most obvious losses are those caused by the feeding of insects

• But it is now known that some of the most serious plant diseases are carried by them.

• The organisms that cause these, whether they are

• Filterable viruses

• Fungi

• Bacteria,

Q: Define these

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Terminologies :

A. Vector
   a. An agent that transports a microorganism from one host to another. Insects that Transmit plant disease are called vectors.

B. Disease
   a. Any condition that impairs the normal functioning of an organism or body. Or Disorder of normal functioning of an organism

C. Epidemiology

a. The study of disease in populations of an organism.

65_66_67_ Insect damages

Q: Define Transmission??and also write it's types??

Transmission is the passing of a pathogen causing communicable disease from an infected Host individual or group to a particular individual or group .Regardless of whether the other Individual was previously infected

Types of transmission

Physical Transmission

In case of mechanical transmission the pathogen is simply carried externally or internally by The insect .Mostly Caused insects that have biting/chewing mouthparts.

Biological Transmission

Specific insect and the specific viral pathogen have some kind of association or relationship Between the two .Mostly Caused insects that have sucking mouthparts

1-The pathogens may enter the plant through feeding injuries made by either the chewing or The sucking type of insect

2-Through punctures made for egg laying and by way of the burrows and galleries of Insects in wood

Particularly in the case of the filterable viruses, those little-known agents causing the so-called Mosaics and related plant diseases .insects become carriers of the disease from plant to Plant.

☘

1. In such cases the insect involved is usually of the piercing-sucking type
2. Often this insect is essential to the overwintering of the disease, sometimes even Necessary to carry the disease from one plant to another Insect Vector born Diseases in Human Another field in which insects are highly injurious is that of human and animal Health and comfort .Some insects, notably mosquitoes and flies, annoy man and

Other animals by their bites and stings .Other insects such as lice have become

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Adapted to more or less continuous life upon animals and cause much discomfort And loss of vitality

Not only do insects live externally on animals but also internally in the skin tissues, and

In a few cases within the flesh and in the alimentary tract

Besides the mental and bodily loss due to their presence, insects transmit disease

Organisms from one person or animal to another

Just as in the transmission of plant diseases the insect may be an accidental agent or An essential carrier

Q: write a note on the insects Borne Diseases??

A few of the more important insect-borne disease in organisms are

- Malaria
- Bubonic plague
- Typhoid
- Typhus
- Yellow fevers.

3. Allergy (hay fever, asthma, etc.) are frequently caused by exposure of susceptible Persons to dried insect scales and hairs
4. Not only is the loss of life because of insect-borne diseases a very serious matter but The total loss of productive labor through illness is tremendous

68-69_ insects benefits

The benefits to man from their activities are frequently overlooked in summing up the Economic importance of insects

Although in many cases it is impossible to say how much good is derived from insects

1-Attack on injurious species

Reliable statistics are easily obtained relative to the production of useful insect products

Those insects which attack injurious species probably form the most important group

They are divided into two sections:

1-The parasites that live in or on the different stages of other insects, then called hosts

2-The predators that capture and devour other insects

Parasites usually, though not always, live in or on a single host insect during their entire Development

- *Whereas a single predator may derive its nourishment from many victim insects.*
- *Parasites are of many species, frequently so specialized in their habits that one kind of*
- *Parasite attacks only one or a few host species*
- *In some cases the parasites work only in insect eggs and thus prevent any feeding loss By the host insect*
- *Other parasites attack early stages of the pests, in some instances killing them soon*
- *In others not until the pests have completed their feeding in spite of being parasitized.*
- *In the latter the benefit is less, but at least there are fewer adults to produce the next Generation*
- *All stages are known to be attacked in different species by one parasite or another*

70- Insect benefits

Q: Define the term Pollination ??

Some plants, as the Smyrna fig, clovers, melons, tomatoes, beans, peas and most of
The common fruits, require the visits of insects before the seeds or fruits can form
Another **highly important group of insects i**ncludes those which pollinate flowers
During their visits for nectar and pollen, thus enabling the plants to produce seeds
The value of these crops which require the services of insects in the United States
Amounts to over two billion dollars

Q: how insects are beneficial in

3-Commercial Products??

A third group of insects is responsible for a number of commercial products

1-Silk

2-Honey

3-Beeswax

4-Shellac

Q how insects acts as a

4-Diet of certain birds, mammals??

Insects provide the entire diet of certain birds, mammals, etc., and an important part

Of the diet for many others

Q: how insects

5-Serve as Food

Even man in some parts of the world consumes large quantities of grasshoppers,

Crickets, ants and other insects

The great value of some insects and the great losses, sometimes enormous, caused

By others make it obviously important to encourage the beneficial forms while striving

To prevent or at least to reduce the losses caused by the injurious ones

70- Insect pest management

Q:define pest management?

The study of principles and experimental approaches to the science that underpins the development of working integrated pest management systems

Q: Types of insect control??

A-NATURAL CONTROL

B-ARTIFICIAL CONTROL:

Measures for the artificial control of insects may be

divided into six groups:

(1) mechanical control

(2) cultural control

(3) biological control,

(4) legislative control

(5) physical control

(6) chemical control.

Regards….🙊

MARYAM

Group links:

For VU expert zoologist Zoo506 https://chat.whatsapp.com/GSEXPT33r6G13leihYgaCD

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" The Expert Zoologist"